# ORBIT CLOSURES AND INVARIANT MEASURES FOR THE REAL REL FLOW IN GENUS TWO

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Abstract. This work studies the dynamics of the real Rel flow from both the topological and measure-theoretic perspectives. We use the interaction between the real Rel flow and tremor deformations to construct examples of orbit closures and invariant measures. In the stratum of genus-2 translation surfaces with two singularities, we show that there exist real Rel orbit closures with no affine structure, as well as ergodic invariant measures for the real Rel flow with generic trajectories lying outside their supports.

## 1. Introduction

This paper considers the stratum $\mathcal{H}(1,1)$, which consists of area-1 translation surfaces of genus 2 with two distinguished singularities. We study the dynamics of real Rel trajectories. That is, given $M \in \mathcal{H}(1,1)$, there is a family of surfaces

$$\mathrm{Rel}_t M \in \mathcal{H}(1,1)$$

obtained by a surgery on $M$ in which the first singularity is fixed and the second singularity is displaced in the horizontal direction, without affecting the geometry away from the singularities. We give a precise definition in Section 3.

We address two types of problems.

**Problem 1.** What can be said about the orbit closures for a given $M \in \mathcal{H}(1,1)$? More precisely, can we describe the sets

$$\overline{\{\mathrm{Rel}_t M : t \geq 0\}}\,?$$

Let $f \in C_c(\mathcal{H}(1,1))$ be a continuous function with compact support and let $M \in \mathcal{H}(1,1)$. Define a family of probability measures $\eta_t^M$ by

$$\int f \; \mathrm{d}\eta_t^M := \frac{1}{t}\int_0^t f(\mathrm{Rel}_s M)\; \mathrm{d}s.$$

**Problem 2.** What can be said about the accumulation points of the family of measures $\{\eta_t^M : t \geq 0\}$ as $t \to \infty$?

### 1.1. Context and motivation.

The real Rel flow is defined on strata of translation surfaces with at least two singularities. It is a one-dimensional subfoliation of the Rel foliation, which has appeared in the literature under various names, including the isoperiodic, kernel and absolute period foliations; see, for example, [BSW22, CDF23, Ham18, McM07b, McM13b, McM14, MW14] and the references therein.

The real Rel flow provides a class of trajectories that are transverse to the $\mathrm{SL}(2,\mathbb{R})$-action. While the $\mathrm{SL}(2,\mathbb{R})$-action has been extensively studied and exhibits strong rigidity properties, much less is understood about the dynamics within Rel leaves. In particular, orbit closures and invariant measures for real Rel trajectories can display a wide range of behaviors, ranging from periodic trajectories to equidistribution phenomena. Understanding these trajectories is closely related to the classification of orbit closures and invariant measures for the horocycle flow [BSW22, CW10]. The questions posed in Problems 1 and 2 can be viewed as Rel analogues of classical problems for homogeneous flows, but in a setting where the dynamics are not governed by a group action.

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1.2. **Statement of results.** Our first main result provides a new answer to Problem 1. It is known that there exist both closed and dense trajectories in every stratum, as well as in certain $\mathrm{SL}(2,\mathbb{R})$-orbit closures; see, for instance, [CW24, HW18, Win22]. Our examples show that there are real Rel orbit closures that do not carry an obvious flat structure and are not invariant under the $\mathrm{SL}(2,\mathbb{R})$-action. In particular, they are not dense in the stratum $\mathcal{H}(1,1)$ nor in any $\mathrm{SL}(2,\mathbb{R})$-orbit closure.

**Theorem 1.1.** *Let $M \in \mathcal{E}(T)$ be a translation surface whose horizontal flow is minimal but not uniquely ergodic. Let $\beta \in \mathcal{T}_M$ be a foliation cocycle. Then there exist $a' \geq 0$ and a modular fiber $\mathcal{E}(T')$ such that*

$$\overline{\{\mathrm{Rel}_t \operatorname{trem}_\beta M : t \in \mathbb{R}\}} = \left\{\operatorname{trem}_{\beta'} M' : M' \in \mathcal{E}(T'),\ \beta' \in \mathcal{T}^{(0)}_{M'},\ |L|_{M'}(\beta') \leq a'\right\}. \tag{1}$$

*The foliation cocycle $\beta$ can be chosen so that the resulting orbit closure is neither equal to $\mathcal{E}(T')$ nor $\mathrm{SL}(2,\mathbb{R})$-invariant.*

The proof of this theorem is presented in Section 6.1. We denote by $\mathcal{H}^{(1/2)}(0)$ the space of area-$\frac{1}{2}$ tori. For $T \in \mathcal{H}^{(1/2)}(0)$, let $\mathcal{E}(T) \subset \mathcal{H}(1,1)$ denote the space of translation surfaces that are double covers of $T$ branched over two points. These are special cases of modular fibers $\mathcal{F}_d(T)$ with $d = 2$. The deformation $\operatorname{trem}_\beta M$ denotes the geometric deformation of the surface $M$ obtained by applying a tremor in the direction of the foliation cocycle $\beta$. These deformations were introduced in [CSW25] in the study of the dynamics of the unipotent group

$$\left\{u_s := \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} : s \in \mathbb{R}\right\}.$$

Foliation cocycles form a subspace $\mathcal{T}_M$ of the tangent space at $M$ in $\mathcal{H}(1,1)$. A distinguished subspace is the space of balanced foliation cocycles $\mathcal{T}^{(0)}_M \subset \mathcal{T}_M$, consisting of those cocycles with zero total variation, that is, $L_M(\beta) = 0$. A related quantity is the total mass $|L|_M(\beta)$, which controls the size of the deformation between $M$ and $\operatorname{trem}_\beta M$.

In Proposition 6.1, which is more general than Theorem 1.1, we give a nearly complete classification of real Rel orbit closures for tremored surfaces in modular fibers with $d = 2$. In [CSW25, Theorem 1.8], the authors show that for every $a > 0$, there exist

$$M \in \mathcal{E} := \bigcup_{T \in \mathcal{H}^{(1/2)}} \mathcal{E}(T) \quad \text{and} \quad \beta \in \mathcal{T}_M,$$

such that

$$\overline{\{u_s \operatorname{trem}_\beta M : s \in \mathbb{R}\}} = \{\operatorname{trem}_{\beta'} M' : M' \in \mathcal{E},\ \beta' \in \mathcal{T}_{M'},\ |L|_{M'}(\beta') \leq a\} =: \mathcal{SF}_{\leq a}.$$

Their proof is non-constructive and uses the Baire category theorem to establish the existence of such surfaces. In contrast, Proposition 6.1 (and Theorem 1.1) shows that the real Rel orbit closure of a tremored surface is *always* of the form given in eq. (1).

Theorem 1.1 strengthens previous examples of orbit closures in that it is constructive and provides an explicit description of the real Rel orbit closure for any surface of the form $\operatorname{trem}_\beta M$, where $M \in \mathcal{E}^{(m\text{-}nue)}(T) \neq \emptyset$. Theorem 1.1 follows from Corollary 6.2 where the corresponding torus $T'$ and constant $a'$ are explicitly identified; see Section 6.1. The existence of tori $T$ for which $\mathcal{E}^{(m\text{-}nue)}(T) \neq \emptyset$ is discussed in Section 3.2, following a result of W. Veech [Vee69]; see Theorem 3.4. In [HW18], it was shown that there exists a real Rel trajectory that is defined for all times and diverges. Consequently, [CW24] observed that it is not clear whether a complete classification of all orbit closures arising in Problem 1 can be achieved. This flexibility has also led to new phenomena and questions. For instance, it was shown in [Osp24] that there exist real Rel trajectories which are neither recurrent nor divergent. In conclusion, Theorem 1.1 shows that, despite this flexibility, the real Rel flow admits a rich and explicitly describable family of orbit closures.

The only previously known answers to Problem 2 arise from generic properties, via the ergodicity results of [CW24]. In contrast, the second main result of this paper provides new, explicit and non-generic examples.

**Theorem 1.2.** *Let $T$ be a torus with no horizontal closed geodesics and let $M \in \mathcal{F}_d(T)$ be a translation surface whose real Rel trajectory is defined for all positive times. Then, for every $\beta \in \mathcal{T}_M$, the weak-$*$ limit of the measures $\{\eta_t^{\operatorname{trem}_\beta M}\}$ exists and satisfies*

$$\eta_t^{\operatorname{trem}_\beta M} \longrightarrow \mu_{T'}, \quad \text{as } t \longrightarrow \infty, \tag{2}$$

*where $\mu_{T'}$ is the probability measure supported on a modular fiber $\mathcal{F}_d(T')$, for some torus $T'$ depending on $M$ and $\beta$. If $\beta \in \mathcal{T}_M^{(0)}$, then $T' = T$.*

*If the horizontal flow on $M$ is not ergodic, $\beta$ can be chosen so the support of $\mu_{T'}$ does not contain the trajectory*

$$\{\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M : t \geq 0\}.$$

The proof of this theorem is presented in Section 5.2. The sets $\mathcal{F}_d(T) \subset \mathcal{H}(1,1)$, called modular fibers, consist of translation surfaces that are $d$-fold branched covers of a torus $T \in \mathcal{H}^{(1/d)}(0)$; see Section 5 for details. These fibers are leaves of the Rel foliation. Analogously, $\mathcal{H}^{(1/d)}(0)$ denotes the space of area-$\frac{1}{d}$ tori.

The measure $\mu_T$ on $\mathcal{F}_d(T)$ is natural in the sense that the Rel foliation induces a flat structure. By the work [Sch05], modular fibers can be viewed as examples of Veech surfaces. The real Rel trajectories on $\mathcal{F}_d(T)$ can therefore be interpreted as horizontal straight-line trajectories on a Veech surface. In particular, such trajectories (when defined for all times) are either periodic or dense and equidistribute with respect to the unique invariant probability measure $\mu_T$.

The novelty of Theorem 1.2 lies in the fact that *every* real Rel trajectory of a surface of the form $\mathrm{trem}_\beta\, M$ equidistributes to some invariant probability measure. Moreover, there exist surfaces $M \in \mathcal{F}_d(T)$ and nonzero foliation cocycles $\beta \in \mathcal{T}_M$ such that, for all $t \in \mathbb{R}$ and for every torus $T'$,

$$\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M \notin \mathcal{F}_d(T').$$

In particular, the trajectory equidistributes to a measure whose support does not contain the trajectory itself.

Theorem 1.2 is closely related to [CSW25, Theorem 1.3]. In that result, the authors consider the space

$$\mathcal{E} = \bigcup_{T \in \mathcal{H}^{(1/2)}} \mathcal{F}_2(T)$$

and the unipotent flow $u_s$, whereas in our setting we replace $\mathcal{E}$ by a fixed modular fiber $\mathcal{F}_d(T)$ and $u_s$ by the real Rel flow $\mathrm{Rel}_t$. The set $\mathcal{E}$ coincides with the locus of area-1 eigenforms of discriminant 4, see [McM07a] for the definition of eigenforms of discriminant $D$. More generally, the sets

$$\mathcal{M}_d(1,1) := \bigcup_{T \in \mathcal{H}^{(1/d)}(0)} \mathcal{F}_d(T)$$

are closed and $\mathrm{SL}(2,\mathbb{R})$-invariant and each connected component supports an $\mathrm{SL}(2,\mathbb{R})$-invariant ergodic probability measure $\mu$, see for instance the results in [EMS03]. The space $\mathcal{M}_d(1,1)$ can be identified with the locus of area-1 eigenforms of discriminant $d^2$, in the terminology of [McM07a].

By [CW24, Theorem 1.2, item (2)], applied with $\mathcal{L} = \mathcal{M}_d(1,1)$, for $\mu$-almost every $M \in \mathcal{M}_d(1,1)$ one has

$$\overline{\{\mathrm{Rel}_t\, M : t \in \mathbb{R}\}} = \mathcal{F}_d(T). \tag{3}$$

Theorem 1.2 provides a non-generic and more explicit description: instead of a $\mu$-almost everywhere statement on $\mathcal{M}_d(1,1)$, we obtain a result that depends *only on the geometry of the base torus.* More precisely, for every torus $T$ with no horizontal closed geodesics and every $M \in \mathcal{F}_d(T)$ (taking $\beta = 0$), if $\mathrm{Rel}_t\, M$ is defined for all $t \geq 0$, then eq. (3) holds.

1.3. **Ideas of the proofs.** To address Problem 1, we begin by analyzing a class of skew products introduced in [Vee69], where conditions for minimality and non-unique ergodicity are established. We relate these skew products to surfaces in $\mathcal{E}(T)$ and use this connection to study the dynamics of real Rel trajectories.

A key ingredient in our approach is the *area exchange mechanism*, which allows us to track the evolution of period coordinates along trajectories of the form $\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M$. Although this technique was introduced in [CSW25] in the study of horocycle trajectories and was first applied to real Rel dynamics in [Osp24], we develop it further to obtain more precise control and explicit calculations; see Section 6.3. The main advantage of this approach is that it replaces non-constructive existence arguments with explicit and quantitative constructions.

For Problem 2, a central new ingredient is Theorem 5.4, which asserts that for any torus $T$ with no horizontal closed geodesics, $\mu_T$-almost every surface $M \in \mathcal{F}_d(T)$ has minimal and uniquely ergodic horizontal flow. This result is a crucial step in the proof of Theorem 1.2 (as well as Proposition 5.1) and allows us to adapt the arguments of [CSW25, Theorem 1.3] to our setting. More generally, the $\mathrm{SL}(2,\mathbb{R})$-invariant ergodic probability measure $\mu$ on $\mathcal{M}_d(1,1)$ disintegrates along modular fibers $\mathcal{F}_d(T)$. While minimality and unique ergodicity are $\mu$-generic, it does not follow that these properties hold with full measure with respect to $\mu_T$ for every torus $T$. Establishing this fiberwise statement is one of the main technical contributions of the paper.

1.4. **Open problems and further directions.** We do not analyze the finer structure of the closures in eq. (2) of Theorem 1.1. In [CSW25, Theorem 1.9], it is shown that the set $\mathcal{SF}_{\le a}$ has Hausdorff dimension in the interval $(5.5, 6)$. This motivates the following question.

**Problem 3.** Let $T$ be a torus with no horizontal closed geodesics such that

$$\mathcal{E}^{(m\text{-}nue)}(T) \neq \emptyset.$$

For every $a > 0$, what can be said about the Hausdorff dimension of the set

$$\mathcal{SU}_{\le a}(T) = \left\{ \mathrm{trem}_{\beta'} M' : M' \in \mathcal{E}(T),\ \beta' \in \mathcal{T}^{(0)}_{M'},\ |L|_{M'}(\beta') \le a \right\}?$$

In particular, does it have non-integer Hausdorff dimension satisfying

$$2 < \mathrm{Hdim}\big(\mathcal{SU}_{\le a}(T)\big) < 3?$$

**Problem 4.** Let $T$ be a torus with no horizontal closed geodesics such that

$$\mathcal{E}^{(m\text{-}nue)}(T) = \emptyset,$$

that is, every surface with minimal horizontal flow is necessarily uniquely ergodic. This regime exhibits different behavior; see, for instance, [Osp24, Theorem 1]. Denote by $\mathcal{E}^{(tor)}(T) \subset \mathcal{E}(T)$ the collection of surfaces whose horizontal straight-line flow decomposes into exactly two minimal components. For $M \in \mathcal{E}^{(tor)}(T)$ and $\beta \in \mathcal{T}_M$, what can be said about the orbit closure

$$\overline{\{\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M : t \in \mathbb{R}\}}?$$

By [CSW25], the $\mathrm{SL}(2,\mathbb{R})$-invariant measure on loci of eigenform of non-squared discriminant is an ergodic measure for the real Rel flow. This provides generic answers to Problems 1 and 2. This motivates the following.

**Problem 5.** Let $M \in \mathcal{H}(1,1)$ belong to a locus of eigenforms whose discriminant is not the square of an integer. What can be said about Problems 1 and 2? In particular, are there analogues of Theorems 1.1 and 1.2 in this setting? Are there surfaces whose horizontal flow is minimal but not uniquely ergodic?

In genus 2, the spaces $\mathcal{M}_d(1,1)$ provide examples of rank 1 loci. In higher genus, the ergodicity of the real Rel flow on rank 1 loci was established by [CW24] and the Rel leaves admit a topological classification by [Ygo22, Theorems A and B].

**Problem 6.** Are there analogues of Theorems 1.1 and 1.2 for surfaces in rank 1 loci and genus greater than 2? Do there exist nontrivial foliation cocycles for surfaces in these loci?

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## 2. Preliminaries

For simplicity, we only cover the definitions for $\mathcal{H}(1,1)$. But most of the following definitions are considered in other strata. We follow the ideas from [CSW25, CW24, MW14, BSW22, AM24] and the references there in. By a surface $S$, we mean a closed, connected, compact surface. Fix $S$ a genus-2 surface and two distinct labeled points $x_1, x_2 \in S$.

2.1. **Translation atlas.** A translation atlas on $S$ with singular points $\Sigma = \{x_1, x_2\}$ of order 1 is a collection

$$\{(U_i, \varphi_i)\}_{i\in\Theta} \sqcup \{(V_k, \psi_k)\}_{k\in\{1,2\}}$$

such that:

- The sets $U_i$ and $V_k$ are open subsets of $S$ and

$$\bigcup_{i\in\Theta} U_i \ \cup\ V_1 \ \cup\ V_2 = S.$$

- For each $i \in \Theta$, the set $U_i$ is disjoint from $\Sigma$ and
$$\varphi_i : U_i \to \mathbb{R}^2$$
is a homeomorphism onto its image.
- The sets $V_1$ and $V_2$ are disjoint neighborhoods of $x_1$ and $x_2$, respectively.
- For $k = 1, 2$,
$$\psi_k : V_k \to \mathbb{R}^2$$
is a 2-fold branched cover onto its image, branched at $x_k$.
- For all $i, j \in \Theta$, the transition maps
$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \mathbb{R}^2$$
are translations of the form
$$w \longmapsto w + c_{ij}.$$
- For all $i \in \Theta$ and $k \in \{1, 2\}$ with $U_i \cap V_k \neq \emptyset$, the transition maps
$$\psi_k \circ \varphi_i^{-1} : \varphi_i(U_i \cap V_k) \longrightarrow \mathbb{R}^2$$
are translations of the form
$$w \longmapsto w + d_{ik}.$$

We write
$$M = (S, \Sigma, \mathcal{A})$$
to represent a translation atlas on $S$ with singularities $\Sigma = \{x_1, x_2\}$. The singularities on $M$ are the distinguished points $\Sigma \subset S$.

Such atlases arise by gluing parallel sides of polygons via translations and endowing the surface with the induced flat geometry. Conversely, every translation atlas on $(S, \Sigma)$ admits a polygonal representation via a decomposition given by the Delaunay triangulation; see [AM24] and references therein.

2.1.1. *Delaunay triangulation.* A Delaunay triangulation of $M$ is a cell decomposition such that:

- The 0-cells are the singular points $x_1, x_2$.
- The 1-cells are straight-line segments in the charts. More precisely, if $\gamma$ is a 1-cell, then for every ordinary chart $(U_i, \varphi_i)$ and every singular chart $(V_k, \psi_k)$ with
$$\gamma \cap U_i \neq \emptyset \qquad \text{or} \qquad \gamma \cap V_k \neq \emptyset,$$
the images
$$\varphi_i(\gamma \cap U_i) \subset \mathbb{R}^2 \qquad \text{and} \qquad \psi_k(\gamma \cap V_k) \subset \mathbb{R}^2$$
are straight-line segments.
- The 2-cells are convex connected 2-simplices in the charts.

2.1.2. *Polygonal representation.* Since $M$ is given by a translation atlas, a polygonal representation of $M$ is a collection of polygons $\{P_\Delta\}$ such that each $P_\Delta$ is obtained as the image of a 2-cell under the charts $(U_i, \varphi_i)$, where the 2-cells arise from the Delaunay triangulation of $M$.

2.1.3. *Translation equivalence and translation surfaces.* Let
$$M_j = \big(S, \Sigma, \{(U_i^{(j)}, \varphi_i^{(j)})\}_{i \in \Theta} \sqcup \{(V_k^{(j)}, \psi_k^{(j)})\}_{k \in \{1,2\}}\big), \quad j = 1, 2,$$
be two translation atlases on $S$ with singularities $\Sigma = \{x_1, x_2\}$.

A map $h : S \to S$ is called a *translation equivalence* between $M_1$ and $M_2$ if:

- $h$ is a homeomorphism,
- $h(x_k) = x_k$ for $k = 1, 2$,
- for every pair of charts $(U_i^{(1)}, \varphi_i^{(1)})$ and $(U_j^{(2)}, \varphi_j^{(2)})$ with
$$h(U_i^{(1)}) \cap U_j^{(2)} \neq \emptyset,$$
the map
$$\varphi_j^{(2)} \circ h \circ (\varphi_i^{(1)})^{-1} : \varphi_i^{(1)}\big(U_i^{(1)} \cap h^{-1}(U_j^{(2)})\big) \longrightarrow \mathbb{R}^2$$
is a translation of the form $w \mapsto w + c$,

- for every pair of charts $(U_i^{(1)}, \varphi_i^{(1)})$ and $(V_k^{(2)}, \psi_k^{(2)})$ with
$$h(U_i^{(1)}) \cap V_k^{(2)} \neq \emptyset,$$
the map
$$\psi_k^{(2)} \circ h \circ (\varphi_i^{(1)})^{-1} : \varphi_i^{(1)}\big(U_i^{(1)} \cap h^{-1}(V_k^{(2)})\big) \longrightarrow \mathbb{R}^2$$
is a translation,
- for every pair of charts $(V_k^{(1)}, \psi_k^{(1)})$ and $(V_\ell^{(2)}, \psi_\ell^{(2)})$ with
$$h(V_k^{(1)}) \cap V_\ell^{(2)} \neq \emptyset,$$
the map
$$\psi_\ell^{(2)} \circ h \circ (\psi_k^{(1)})^{-1} : \psi_k^{(1)}\big(V_k^{(1)} \cap h^{-1}(V_\ell^{(2)})\big) \longrightarrow \mathbb{R}^2$$
is a translation.

Translation equivalence maps define an equivalence relation on translation atlases.

A *translation surface* is an equivalence class of translation atlases on $(S, \Sigma)$ under translation equivalence maps. We denote a translation surface by $M$ for simplicity, since it is enough to specify either a polygonal representation or a translation atlas on $(S, \Sigma)$. A translation surface has a family of flows $\{\Upsilon_t^\theta\}_{\theta \in [0,2\pi)}$. These flows are given by straight-line trajectories in local coordinates in direction $\theta$, and satisfy the relations
$$\Upsilon_{t+s}^\theta = \Upsilon_t^\theta \circ \Upsilon_s^\theta \quad \text{and} \quad \Upsilon_{-t}^\theta = \Upsilon_t^{\theta+\pi}.$$
Each translation surface comes equipped with a natural Lebesgue measure Leb, obtained by pulling back the Lebesgue measure on $\mathbb{R}^2$. In particular,
$$\text{area}(M) = \text{Leb}(M).$$
This measure is invariant under the flow $\Upsilon_t^\theta$. It was shown in [KMS86] that, for every translation surface $M$, the measure Leb is ergodic with respect to $\Upsilon_t^\theta$ for almost every $\theta \in [0, 2\pi)$.

2.2. **Connected sums of tori.** Denote by $\mathcal{H}(0)$ the collection of flat tori of area 1, which can be identified with
$$\mathrm{SL}(2,\mathbb{R})/\,\mathrm{SL}(2,\mathbb{Z}).$$
A torus $T \in \mathcal{H}(0)$ can be written as $\mathbb{R}^2/\Lambda$, where $\Lambda$ is a lattice in $\mathbb{R}^2$ of covolume 1.

The notation $\mathcal{H}(0)$ indicates that we mark a point in $\mathbb{R}^2/\Lambda$, for instance the class of $(0, 0)$. Each $T \in \mathcal{H}(0)$ has an atlas
$$\{(U_i, \varphi_i)\}_{i \in I}$$
given by the quotient map
$$\mathbb{R}^2 \longrightarrow T.$$
Here each
$$\varphi_i : U_i \longrightarrow \mathbb{R}^2$$
is a homeomorphism onto its image and, for $U_i \cap U_j \neq \emptyset$, the transition maps
$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j)$$
are translations of the form
$$z \longmapsto z + z_{ij}.$$
Similarly, we can define translation equivalence maps.

A segment $I$ on $T_1$ and $T_2$ have the same length and direction if there are embeddings
$$I_1 : [a, b] \longrightarrow T_1, \qquad I_2 : [a, b] \longrightarrow T_2,$$
which, in charts, are straight-line segments starting at the marked point, have length $b - a$ and have the same direction.

The connected sum
$$T_1 \#_I T_2$$
is the topological connected sum of $T_1$ and $T_2$ along the segment $I$. Opposite sides of the embedded segments
$$J_1 \subset T_1, \qquad J_2 \subset T_2$$
are identified.

The resulting topological surface has genus 2 and has two marked points $x_1, x_2$. By convention, $x_1$ is the identification of the marked points of $T_1$ and $T_2$. The point $x_2$ is the other endpoint of the embedded segment $I$.

The translation atlas away from the singularities is given by the translation atlases on $T_1$ and $T_2$. For the singularities $x_1$ and $x_2$, a neighborhood $V_i$ can be taken to be of the form

$$V_i = (D_1 \sqcup D_2)/\sim,$$

where $D_1 \subset T_1$ and $D_2 \subset T_2$ are small disks of the same radius centered at $x_i$ and $\sim$ identifies opposite sides of the removed portion of $I$.

By taking the radius small enough, we may assume that

$$V_1 \cap V_2 = \emptyset.$$

The chart

$$\psi_i : V_i \longrightarrow \mathbb{R}^2$$

is a branched 2-fold cover onto a disk in $\mathbb{R}^2$. For points on $I$ that are not singular, the structure is obtained by identifying two embedded half-disks in $T_1$ and $T_2$ along their diameters, each half lying in one of the tori. The resulting translation surface is also denoted

$$M = T_1 \#_I T_2.$$

Note that $T_2 \#_I T_1$ and $T_1 \#_I T_2$ are translation equivalent. Indeed, the connected sum is topologically commutative, and the natural homeomorphism exchanging the two tori can be chosen explicitly so that it is a translation in charts.

In [McM07a], it was proved that *every* translation surface of genus 2 with two singularities of order 1 can be represented as a connected sum of two tori.

Order 1 means that there is an extra $2\pi$ angle around each singularity $x_i$. In other words, since a neighborhood around each singularity is a branched cover of a disk, the total angle around $x_i$ is

$$2\pi(k+1) = 4\pi.$$

2.3. **The strata of translation surfaces.** We denote by $\widehat{\mathcal{H}}(1,1)$ the collection of translation surfaces of genus 2 with two distinguished singularities of order 1. Each translation surface has positive area. By $\mathcal{H}^{(a)}(1,1)$ we mean translation surfaces in $\widehat{\mathcal{H}}(1,1)$ of area $a$ and for area $a = 1$ we write

$$\mathcal{H}(1,1) := \mathcal{H}^{(1)}(1,1).$$

The area of a translation surface is the sum of the areas of the polygons in a polygonal representation. Given $M \in \widehat{\mathcal{H}}(1,1)$, we write $M = (S, \{x_1, x_2\}, \mathcal{A})$, where $\mathcal{A}$ is a translation atlas; we regard $\mathcal{A}$ as an equivalence class under translation equivalence maps.

Let $(S', \{x_1', x_2'\})$ be a genus 2 surface with two distinguished points $x_1'$ and $x_2'$, called the model surface. A pair $(M, f)$ is a *marking* of a translation surface $M$ if $f$ is a homeomorphism $f : S' \longrightarrow S$ that preserves the labeled points:

$$f(x_i') = x_i, \quad i = 1, 2.$$

Two markings $(M, f_1)$ and $(M, f_2)$ are said to be equivalent if there exists a translation equivalence map

$$h : M \longrightarrow M$$

such that $h \circ f_1$ is isotopic to $f_2$ via an isotopy that fixes $\{x_1', x_2'\}$ pointwise. The collection of equivalence classes of marked translation surfaces defines the stratum $\widehat{\mathcal{H}}_m(1,1)$ and the collection of equivalence classes of markings of translation surfaces of area $a$ is denoted by $\mathcal{H}_m^{(a)}(1,1)$. When $a = 1$, we write $\mathcal{H}_m(1,1)$. Without loss of generality, we assume that $S' = S$ and $x_i' = x_i$ for $i = 1, 2$. Since marked translation surfaces are classes $[(M, f)]$, to avoid using this notation we simply write $(M, f)$ for the class and for the representative.

2.4. **Manifold structure of marked strata.** A translation surface $M \in \widehat{\mathcal{H}}(1,1)$ is equipped with two real 1-forms $\mathrm{d}x$ and $\mathrm{d}y$ by pulling back the standard forms on $\mathbb{R}^2$ with the translation atlas. Since the transition maps are translations and representatives of the atlas $\mathcal{A}$ differ by translation equivalence maps, these 1-forms are well-defined. These forms define cocycles

$$\mathrm{hol}_M^{(x)},\ \mathrm{hol}_M^{(y)} \in H^1(M,\Sigma;\mathbb{R}),$$

obtained by integrating along oriented curves $\gamma$ representing homology classes $[\gamma] \in H_1(M,\Sigma;\mathbb{Z})$:

$$\mathrm{hol}_M^{(x)}([\gamma]) = \int_\gamma \mathrm{d}x, \qquad \mathrm{hol}_M^{(y)}([\gamma]) = \int_\gamma \mathrm{d}y.$$

We denote by $\mathrm{hol}_M := (\mathrm{hol}_M^{(x)}, \mathrm{hol}_M^{(y)})$, the cocycle $\mathrm{hol}_M \in H^1(M,\Sigma;\mathbb{R}^2)$ using the identification

$$H^1(M,\Sigma;\mathbb{R}^2) \simeq H^1(M,\Sigma;\mathbb{R}) \oplus H^1(M,\Sigma;\mathbb{R}).$$

Let

$$\mathrm{dev} : \widehat{\mathcal{H}}_m(1,1) \longrightarrow H^1(S,\Sigma;\mathbb{R}^2)$$

be the map that assigns to an equivalence class of a marked surface $(M,f)$ the cocycle

$$f^*\mathrm{hol}_M : [\gamma] \longmapsto \mathrm{hol}_M([f(\gamma)]), \qquad [\gamma] \in H_1(S,\Sigma;\mathbb{Z}).$$

We denote by

$$\mathrm{hol}_{(M,f)} := f^*\mathrm{hol}_M \in H^1(S,\Sigma;\mathbb{R}^2) \quad \text{and} \quad \mathrm{hol}^{(x)}_{(M,f)} := f^*\mathrm{hol}_M^{(x)},\ \mathrm{hol}^{(y)}_{(M,f)} := f^*\mathrm{hol}_M^{(y)} \in H^1(S,\Sigma;\mathbb{R}),$$

the corresponding cocycles.

Given a Delaunay triangulation $\tau$ of $M$, it induces a triangulation of $S$ via the marking, $\tau_f := f^{-1}(\tau)$. For a 1-cell $e$ in this triangulation and a cocycle

$$\beta \in H^1(S,\Sigma;\mathbb{R}^2),$$

the value $\beta(e)$ is a vector in $\mathbb{R}^2$. For a 2-cell $\Delta \in \tau$, $\beta(\Delta)$ is a Euclidean triangle (possibly degenerate) in $\mathbb{R}^2$ whose edges have the orientations given by $\beta(e)$.

We say $\beta$ preserves the orientation of $\Delta = [e_1,e_2,e_3]$ if the determinants

$$\det(\beta(e_1-e_3),\beta(e_2-e_3)) \quad \text{and} \quad \det(\mathrm{hol}_M(e_1-e_3),\mathrm{hol}_M(e_2-e_3))$$

have the same sign. We assume that every 2-cell $\Delta \in \tau$ is non-degenerate, that is, the triangle $\mathrm{hol}_M(\Delta)$ is non-degenerate. Thus, if $\beta$ preserves the orientation of $\Delta$, then $\beta(\Delta)$ is also a non-degenerate Euclidean triangle.

Define

$$U_{f,M,\tau} \subset H^1(S,\Sigma;\mathbb{R}^2)$$

to be the set of cocycles $\beta$ that preserve the orientation for every 2-cell $\Delta \in \tau$. This set $U_{f,M,\tau}$ is open. Given $\beta \in U_{f,M,\tau}$, we can construct a translation surface $M_{\beta,\tau}$ as follows. For each 2-cell $\Delta$, let $\beta(\Delta) \subset \mathbb{R}^2$ be the Euclidean triangle determined by $\beta$. The edges of the triangles $\{\beta(\Delta)\}_{\Delta\in\tau}$ are glued according to the combinatorics of the polygonal representation of $M$ given by the triangles $\{\mathrm{hol}_M(\Delta)\}_{\Delta\in\tau}$. Denote the resulting translation surface by $M_{\beta,\tau}$.

There is a natural piecewise linear map in charts

$$g_{\beta,\tau} : M \longrightarrow M_{\beta,\tau} \tag{4}$$

defined by linear maps sending the Euclidean triangles $\mathrm{hol}_M(\Delta)$ to $\beta(\Delta)$. These maps are determined by the action on the vectors

$$\mathrm{hol}_M(e_1-e_3) \mapsto \beta(e_1-e_3), \qquad \mathrm{hol}_M(e_2-e_3) \mapsto \beta(e_2-e_3)$$

for all $\Delta = [e_1,e_2,e_3] \in \tau$. The map $g_{\beta,\tau}$ is a homeomorphism, since it preserves the combinatorics of the polygonal representations of $M$ and $M_{\beta,\tau}$. Consequently, we obtain the homeomorphism

$$f_{\beta,\tau} := g_{\beta,\tau} \circ f : S \longrightarrow M_{\beta,\tau},$$

and hence the marked translation surface $(M_{\beta,\tau}, f_{\beta,\tau}) \in \widehat{\mathcal{H}}_m(1,1)$.

Let

$$V_{f,M,\tau} \subset \widehat{\mathcal{H}}_m(1,1)$$

be the set of equivalence classes of marked surfaces obtained in this way from $U_{f,M,\tau}$. Define the map

$$\Phi_{f,M,\tau} : U_{f,M,\tau} \longrightarrow V_{f,M,\tau}, \qquad \beta \mapsto (M_{\beta,\tau}, f_{\beta,\tau}).$$

By construction, $\Phi_{f,M,\tau}$ is injective and

$$\operatorname{dev} \circ \Phi_{f,M,\tau} = \operatorname{Id}_{U_{f,M,\tau}},$$

hence dev is injective when we restrict it to $V_{f,M,\tau}$.

We fix a marked translation surface $(M, f)$ and a Delaunay triangulation $\tau$ to define $V_{f,M,\tau}$, $U_{f,M,\tau}$ and $\Phi_{f,M,\tau}$. One checks that these do not depend on the choice of $\tau$, so we write $V_{f,M}$, $U_{f,M}$ and $\Phi_{f,M}$.

The collection $\{V_{f,M}\}$ defines a topology on $\widehat{\mathcal{H}}_m(1,1)$ and the collection of charts $\{(V_{f,M}, \operatorname{dev})\}$ defines an atlas on $\widehat{\mathcal{H}}_m(1,1)$. We call this atlas the *period coordinates*. The topology we defined is the coarsest topology for which each chart map is continuous. The transition maps between charts are linear maps between open subsets of $H^1(S, \Sigma; \mathbb{R}^2)$. Therefore, $\widehat{\mathcal{H}}_m(1,1)$ has the structure of a linear manifold.

The tangent space at $(M, f)$ is identified with $T_{(M,f)}\widehat{\mathcal{H}}_m(1,1) \simeq H^1(S, \Sigma; \mathbb{R}^2)$. The tangent bundle $T\widehat{\mathcal{H}}_m(1,1)$ is identified with $\widehat{\mathcal{H}}_m(1,1) \times H^1(S, \Sigma; \mathbb{R}^2)$.

2.5. **Manifold and orbifold structure.** Let $\operatorname{Mod}(S, \{x_1, x_2\})$ be the mapping class group, that is, the group of isotopy classes of orientation preserving homeomorphisms fixing $\{x_1, x_2\}$ pointwise:

$$h : S \longrightarrow S, \qquad h(x_i) = x_i, \quad i = 1, 2.$$

This group acts on $\widehat{\mathcal{H}}_m(1,1)$ by precomposition of the marking:

$$[h] \cdot (M, f) := (M, f \circ h^{-1}), \qquad [h] \in \operatorname{Mod}(S, \Sigma).$$

This action is continuous, *almost free* and *proper*. Almost free means that for every $(M, f) \in \widehat{\mathcal{H}}_m(1,1)$, the stabilizer

$$\operatorname{Stab}(M, f) := \{[h] \in \operatorname{Mod}(S, \Sigma) : (M, f \circ h) = (M, f)\}$$

is finite. Proper (or properly discontinuous) means that for every compact set $K \subset \widehat{\mathcal{H}}_m(1,1)$, the set

$$\{[h] \in \operatorname{Mod}(S, \Sigma) : [h]K \cap K \neq \emptyset\}$$

is finite.

The group $\operatorname{Mod}(S, \Sigma)$ acts transitively on the fibers

$$\{(M, f) : f : S \to M \text{ is a marking}\}.$$

Therefore, the projection

$$\widehat{\mathcal{H}}_m(1,1) \longrightarrow \widehat{\mathcal{H}}(1,1), \quad (M, f) \longmapsto M$$

equips $\widehat{\mathcal{H}}(1,1)$ with the quotient topology

$$\widehat{\mathcal{H}}(1,1) = \widehat{\mathcal{H}}_m(1,1)/\operatorname{Mod}(S, \Sigma).$$

Moreover, the linear manifold structure of $\widehat{H}_m(1,1)$ equips $\widehat{\mathcal{H}}(1,1)$ with an orbifold structure. We consider the $\operatorname{Mod}(S, \Sigma)$-action on $H^1(S, \Sigma; \mathbb{R}^2)$ by precomposition, that is, if $[h] \in \operatorname{Mod}(S, \Sigma)$ and $\beta \in H^1(S, \Sigma; \mathbb{R}^2)$, then

$$[h]\beta : \gamma \mapsto \beta(h^{-1}(\gamma)).$$

2.5.1. *Stabilizers.* Suppose that $[h] \in \operatorname{Stab}(M, f) \neq \{Id\}$ and $[h] \neq Id$. Then $(M, f)$ and $(M, f \circ h)$ represent the same marked surface. This means there exists a translation equivalence

$$g : M \longrightarrow M$$

such that $f \circ h^{-1}$ and $g \circ f$ are isotopic.

**Example 2.1.** Let $M = T_1 \#_I T_2$ be the translation surface obtained by the connected sum of two tori. Suppose initially that $T_1$ and $T_2$ are the same torus. Thus $M = T \#_I T$ and $I \subset T$ is a geodesic segment. Denote by $\partial I \subset T$ the two endpoints of $I$. A marking can be described by labeling each copy of the torus $T$. Then $T_{\mathrm{A}} \#_I T_{\mathrm{B}}$ and $T_{\mathrm{B}} \#_I T_{\mathrm{A}}$ define two markings of $M$, the map $\iota_M$ that flips the labels

$$\iota_M : (x, \mathrm{A}) \longleftrightarrow (x, \mathrm{B}) \quad x \in T \setminus \{\partial I\}$$

is a translation in charts and fixes the singularities of $M$ pointwise. Hence, the markings $T_{\mathrm{A}} \#_I T_{\mathrm{B}}$ and $T_{\mathrm{B}} \#_I T_{\mathrm{A}}$ belong to the same equivalence class in $\widehat{\mathcal{H}}_m(1,1)$.

Once and for all, for every surface that arises as of two tori glued along a slit, we denote by

$$\iota_M : M \longrightarrow M$$

the involution that exchanges the labels of the copies of the torus $T$ in $M$. The involution $\iota_M$ is nontrivial on absolute homology $H_1(M;\mathbb{Z})$, hence is not isotopic to the identity.

Let $f : S \to M$ be a marking and $\iota : M \to M$ be a translation equivalence map that is not isotopic to the identity. Then $\iota \circ f : S \to M$ is also a marking. By definition, it is immediate that the classes $(M, \iota \circ f)$ and $(M, f)$ are the same

$$(M, \iota \circ f) = (M, f).$$

In deed, the class $[h] = [f^{-1} \circ \iota \circ f] \in \mathrm{Mod}(S, \Sigma)$ acts in a nontrivial way on $H_1(S, \Sigma; \mathbb{Z})$ hence it is not the identity and

$$[h](M, f) = (M, f \circ f^{-1} \circ \iota \circ f) = (M, \iota \circ f) = (M, f).$$

Therefore, the stabilizer $\mathrm{Stab}(M, f)$ is nontrivial.

Let $\iota$ be a nontrivial translation equivalence $\iota : M \to M$ for $M \in \widehat{\mathcal{H}}(1,1)$. For a fixed direction $\theta$, there are exactly two prongs at each singularity. Therefore, any such $\iota$ must permute the prongs, this implies that $\iota^2 = \mathrm{Id}_M$ is the identity map. The quotient map $M \to M/\iota$ is a 2-fold branched covering and since $\iota$ is non-trivial, $M/\iota$ must be homeomorphic to a torus with two ramification points. The quotient map endows $M/\iota$ with a translation atlas, hence $M = M/\iota \#_I M/\iota$ for some geodesic segment $I \subset M/\iota$ connecting the ramification points. This shows that for every $M \in \widehat{H}(1,1)$, the surfaces that are 2-fold branched covers of tori are the only translation surfaces of genus 2 with two singularities that have a non-trivial translation equivalence and in fact one, $\iota_M$.

We denote by

$$\widehat{\mathcal{E}}_m \subset \widehat{\mathcal{H}}_m(1,1)$$

the set of marked translation surfaces with nontrivial stabilizer. Let $\widehat{\mathcal{E}} \subset \widehat{\mathcal{H}}(1,1)$ be its image under the quotient map. When we restrict to area $a$, we denote the corresponding loci by $\mathcal{E}_m^{(a)} \subset \widehat{\mathcal{E}}_m$ and $\mathcal{E}^{(a)} \subset \widehat{\mathcal{E}}$. Likewise, when $a = 1$, we write $\mathcal{E}_m$ and $\mathcal{E}$.

2.5.2. *Orbifold charts.* An orbifold atlas for $\widehat{\mathcal{H}}(1,1)$ is given by charts of the form

$$(U_{M,f}, \Phi_{M,f}, \mathrm{Stab}(M, f))$$

where $\mathrm{Stab}(M, f)$ acts on $H^1(S, \Sigma; \mathbb{R}^2)$ by precomposition on cocycles. If $(M, f) \notin \widehat{\mathcal{E}}_m$, then $\mathrm{Stab}(M, f) = \{\mathrm{Id}\}$. Otherwise,

$$\mathrm{Stab}(M, f) = \{\mathrm{Id}, [f^{-1} \circ \iota_M \circ f]\},$$

where $\iota_M$ corresponds to the involution of $M$ described above.

Let $(M, f) \in \widehat{\mathcal{H}}_m(1,1)$ with $\mathrm{Stab}(M, f) \neq \{\mathrm{Id}\}$. The marked translation surfaces

$$(M_{\beta,\tau}, f_{\beta,\tau}) \in V_{M,f} \quad \text{and} \quad (M_{\beta,\tau}, f_{\beta,\tau} \circ f^{-1} \circ \iota_M \circ f)$$

are the same. This is because

(1) $f_{\beta,\tau} = g_{\beta,\tau} \circ f$ where $g_{\beta,\tau}$ is the map in eq. (4),
(2) the composition of $g_{\beta,\tau} \circ \iota_M \circ g_{\beta,\tau}^{-1}$ and $f_{\beta,\tau} \circ f^{-1} \circ \iota_M \circ f$ is exactly $f_{\beta,\tau}$ and
(3) $g_{\beta,\tau} \circ \iota_M \circ g_{\beta,\tau}^{-1} : M_{\beta,\tau} \to M_{\beta,\tau}$ is a translation equivalence.

Then $\mathrm{Stab}(M, f)U_{M,f} = U_{M,f}$.

If $\beta \in \widehat{\mathcal{H}}_m(1,1)$ represents a vector in the tangent space $T_{(M,f)}H^1(S, \Sigma; \mathbb{R}^2)$ at $(M, f) \in \widehat{\mathcal{E}}_m$, the action of the nontrivial element $[h]$ of the stabilizer identifies $\beta$ and $\beta \circ h$ in the tangent space of $M \in \widehat{\mathcal{H}}(1,1)$. In other words, any deformation $M'$ of $M$ can be studied by choosing either markings $f : S \to M$ and $f' : S \to M'$, or the markings $f : S \to M$ and $f' \circ h : S \to M'$ in order to make a comparison in period coordinates. This will be a recurrent idea in the proofs.

2.5.3. *Dimension.* The manifold $\widehat{\mathcal{H}}_m(1,1)$ has real dimension

$$\dim_{\mathbb{R}} \widehat{\mathcal{H}}_m(1,1) = \dim_{\mathbb{R}} H^1(S, \{x_1, x_2\}; \mathbb{R}^2) = 10$$

and the quotient $\widehat{\mathcal{H}}(1,1)$ has dimension 10 as an orbifold. The area map is a continuous function $\widehat{\mathcal{H}}(1,1) \to \mathbb{R}_{>0}$ and invariant under the action of $\mathrm{Mod}(S,\Sigma)$. Hence the fixed-area strata $\mathcal{H}_m^{(a)}(1,1)$ and $\mathcal{H}^{(a)}(1,1)$ have codimension 1. Thus

$$\dim_{\mathbb{R}} \mathcal{H}_m^{(a)}(1,1) = 9, \qquad \dim_{\mathbb{R}} \mathcal{H}^{(a)}(1,1) = 9.$$

The locus $\widehat{\mathcal{E}}_m \subset \widehat{\mathcal{H}}_m(1,1)$ is a submanifold defined by linear constraints on period coordinates. The restrictions are

$$\mathrm{dev}(M,f)([\gamma_j^1]) = \mathrm{dev}(M,f)([\gamma_j^2]), \quad j = 1,2 \tag{5}$$

where $\{\gamma_j^k\}_{j,k=1,2}$ are closed curves in $S$ generating a symplectic basis for $H_1(S;\mathbb{Z})$ viewed as elements of $H_1(S,\Sigma;\mathbb{Z})$. Thus $\widehat{\mathcal{E}}_m$ has codimension 4 because the restrictions in eq. (5) gives two equations with values in $\mathbb{R}^2$ and hence

$$\dim_{\mathbb{R}} \widehat{\mathcal{E}}_m = \dim_{\mathbb{R}} \widehat{\mathcal{E}} = 6, \qquad \dim_{\mathbb{R}} \mathcal{E}_m^{(a)} = \dim_{\mathbb{R}} \mathcal{E}^{(a)} = 5.$$

2.5.4. *Comparison maps.* Let $M$, $M_{\beta,\tau}$ be translation surfaces and $g_{\beta,\tau}$ be as in the discussion of eq. (4). The map

$$g_{\beta,\tau} : M \longrightarrow M_{\beta,\tau}$$

is piecewise linear in charts and differentiable on the interior of each 2-cell.

To say that $M_j \to M$ as $j \to \infty$ is equivalent to choosing markings $(M_j, f_j)$ and $(M, f)$ such that, for sufficiently large $j$, the comparison maps

$$\phi_j := f \circ f_j^{-1} : M_j \longrightarrow M$$

are piecewise affine in charts and their derivatives wherever defined converge to the identity matrix.

2.6. **Linear action.** Let $\mathrm{SL}(2,\mathbb{R})$ be the group of $2 \times 2$-matrices with determinant 1. We distinguish two particular subgroups:

$$D = \left\{ g_t := \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} : t \in \mathbb{R} \right\} \quad \text{and} \quad U = \left\{ u_s := \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} : s \in \mathbb{R} \right\}.$$

Denote by $DU$ the set $\{g_t u_s : t, s \in \mathbb{R}\}$ obtained by matrix multiplication. Let $A \in \mathrm{SL}(2,\mathbb{R})$ be a matrix and let $M$ be a translation surface. The matrix $A$ acts on $M = (S, \Sigma, \mathcal{A})$ by composition in the charts: if $(U_\alpha, \varphi_\alpha) \in \mathcal{A}$, define $(U_\alpha, A\varphi_\alpha)$. Write $A\mathcal{A} := \{(U_\alpha, A\varphi_\alpha) : (U_\alpha, \varphi_\alpha) \in \mathcal{A}\}$. Then

$$AM := (S, \Sigma, A\mathcal{A}).$$

$\mathrm{SL}(2,\mathbb{R})$ acts on $H^1(S,\Sigma;\mathbb{R}^2)$ as a linear map on the coefficients $\mathbb{R}^2$. In a polygonal representation $\{P_\Delta\}/\sim$ of $M$, we can view $AM$ as the surface obtained from the polygons $\{AP_\Delta\}$, where $A$ acts linearly on the polygons and the gluing is done using the same combinatorics as for $M$. As a consequence, for every $A \in \mathrm{SL}(2,\mathbb{R})$ and $M \in \widehat{\mathcal{H}}(1,1)$, we obtain an *affine comparison homeomorphism*

$$\phi_A : M \longrightarrow AM,$$

since it is a linear map in charts and the derivative is $A$. This map satisfies the equivariant property

$$\mathrm{hol}_{AM}(\phi_A(\gamma)) = A\,\mathrm{hol}_M(\gamma) \quad \text{for every curve} \quad \gamma \subset M.$$

The action on $\widehat{\mathcal{H}}(1,1)$ is continuous and it lifts to a continuous action on $\widehat{\mathcal{H}}_m(1,1)$ by

$$(M,f) \longmapsto (AM, \phi_A \circ f).$$

The $\mathrm{SL}(2,\mathbb{R})$-action has many properties: It is equivariant with respect to the quotient map $\widehat{\mathcal{H}}_m(1,1) \to \widehat{\mathcal{H}}(1,1)$, preserves area so $\mathcal{H}_m^{(a)}(1,1)$ and $\mathcal{H}^{(a)}(1,1)$ are invariant, it is equivariant on the developing map

$$\mathrm{dev}(AM, \phi_A \circ f)) = Af^*\,\mathrm{hol}_M$$

and it leaves invariant the loci $\widehat{\mathcal{E}}_m$ and $\widehat{\mathcal{E}}$.

## 3. Rel foliation and real Rel flow

Let

$$\mathrm{Res} : H^1(S,\Sigma;\mathbb{R}^2) \longrightarrow H^1(S;\mathbb{R}^2)$$

be the restriction map induced by the inclusion $H_1(S;\mathbb{Z}) \hookrightarrow H_1(S,\Sigma;\mathbb{Z})$. Since $|\Sigma| = 2$, the kernel

$$\mathfrak{R} := \mathrm{Ker}(\mathrm{Res})$$

is non-trivial and is a subspace of $H^1(S,\Sigma;\mathbb{R}^2)$ of dimension 2. The cocycles in $\mathfrak{R}$ can be identified with equivalence classes of $\mathbb{R}^2$-valued functions on $\Sigma$, where $\beta_1, \beta_2 : \Sigma \to \mathbb{R}^2$ are equivalent if $\beta_1 - \beta_2$ is constant. See the definitions in [BSW22]. We identify

$$\mathfrak{R} \simeq \mathfrak{R}_x \oplus \mathfrak{R}_y$$

using the identification

$$H^1(S,\Sigma;\mathbb{R}^2) \simeq H^1(S,\Sigma;\mathbb{R}) \oplus H^1(S,\Sigma;\mathbb{R}).$$

The foliation obtained by integrating $\mathfrak{R}$ on $\widehat{\mathcal{H}}_m(1,1)$ is called the full *Rel foliation*. The subspace

$$\mathfrak{R}_x \simeq \mathfrak{R}_x \oplus 0 \subset \mathfrak{R}$$

defines a subfoliation on $\widehat{\mathcal{H}}_m(1,1)$ whose leaves are 1-dimensional. Given $(M,f) \in \widehat{\mathcal{H}}_m(1,1)$, we denote by

$$\{\mathrm{Rel}_t(M,f) : t \in D_{(M,f)}\}$$

the path satisfying the ODE

$$\mathrm{Rel}_0(M,f) = (M,f), \qquad \frac{d}{dt}\,\mathrm{Rel}_t(M,f) = 1 \in \mathfrak{R}_x,$$

with maximal domain $D_{(M,f)} \subset \mathbb{R}$. The cocycle $1 \in \mathfrak{R}_x$ can be interpreted as measuring the signed horizontal displacement from $x_1$ to $x_2$ on $S$ according to the translation structure of $(M,f)$. The map Res is equivariant with respect to the action of $\mathrm{Mod}(S,\Sigma)$, so the paths $\{\mathrm{Rel}_t(M,f)\}$ descend to well-defined paths on $\widehat{\mathcal{H}}(1,1)$.

Let

$$\widehat{\mathcal{R}}_{\infty,m} \subset \widehat{\mathcal{H}}_m \quad \text{and} \quad \widehat{\mathcal{R}}_{\geq 0,m} \subset \widehat{\mathcal{H}}_m$$

be the subsets of surfaces for which $D_{(M,f)} = \mathbb{R}$ and $D_{(M,f)} = [0,\infty)$, respectively. Both $\widehat{\mathcal{R}}_{\infty,m}$ and $\widehat{\mathcal{R}}_{\geq 0,m}$ are open by standard ODE arguments. Thus they map to open sets $\widehat{\mathcal{R}}_\infty$ and $\widehat{\mathcal{R}}_{\geq 0}$ respectively. These sets are open and dense in $\widehat{\mathcal{H}}(1,1)$. We remark that $\widehat{\mathcal{R}}_\infty \subsetneq \widehat{\mathcal{R}}_{\geq 0} \subsetneq \widehat{\mathcal{H}}$. The issue of defining $\mathrm{Rel}_t\, M$ for all $t$ was studied by [MW14], they proved that for $M \in \widehat{\mathcal{H}}(1,1)$, $\mathrm{Rel}_t\, M$ is defined if and only if there are no horizontal saddle connections $\gamma$ in $M$ from $x_1$ to $x_2$ such that $\mathrm{hol}^{(x)}_M(\gamma) = -t$. It was proved in [BSW22] that $\widehat{\mathcal{R}}_\infty$ is dense, which implies that $\widehat{\mathcal{R}}_{\geq 0}$ is also dense. Moreover, surfaces lying in the same Rel leaf have the same area.

We write $\mathrm{Rel}_t\, M$ for the image of $\mathrm{Rel}_t(M,f)$ under the quotient map. In [BSW22, Proposition 4.10], the flow property is established: if $\mathrm{Rel}_{t_1} M$ and $\mathrm{Rel}_{t_2}(\mathrm{Rel}_{t_1} M)$ are defined, then $\mathrm{Rel}_{t_1+t_2} M$ is defined and

$$\mathrm{Rel}_{t_1+t_2} M = \mathrm{Rel}_{t_2}(\mathrm{Rel}_{t_1} M).$$

Moreover, for each $t \in D_{(M,f)}$, the map

$$(M,f) \longmapsto \mathrm{Rel}_t(M,f)$$

is continuous at $(M,f)$. Consequently, the induced map

$$M \longmapsto \mathrm{Rel}_t\, M$$

is continuous at $M$ whenever $\mathrm{Rel}_t\, M$ is defined. With the identification $\mathfrak{R}_x \simeq \mathbb{R}$, we may view motion along the real Rel leaves as a flow called the *real Rel* flow.

For every translation surface $M$, there exists $\varepsilon_M > 0$ such that $\mathrm{Rel}_t\, M$ is defined for all $|t| < \varepsilon_M$. For such $t$, lifting to the marked stratum, we define the *real Rel comparison map*

$$\phi_t : M \longrightarrow \mathrm{Rel}_t\, M$$

using the comparison map of eq. (4) with $\beta = t \cdot 1 \in \mathfrak{R}_x \oplus 0$.

In local coordinates,

$$\mathrm{hol}^{(x)}_{\mathrm{Rel}_t M}(\phi_t(\gamma)) = \mathrm{hol}^{(x)}_M(\gamma) + t, \qquad \mathrm{hol}^{(y)}_{\mathrm{Rel}_t M}(\phi_t(\gamma)) = \mathrm{hol}^{(y)}_M(\gamma)$$

for every saddle connection $\gamma \subset M$ from $x_1$ to $x_2$, and

$$\mathrm{hol}^{(x)}_{\mathrm{Rel}_t M}(\phi_t(\gamma)) = \mathrm{hol}^{(x)}_M(\gamma), \qquad \mathrm{hol}^{(y)}_{\mathrm{Rel}_t M}(\phi_t(\gamma)) = \mathrm{hol}^{(y)}_M(\gamma)$$

for every closed curve $\gamma \subset M$. This map can be extended for all $t$ for which $\mathrm{Rel}_t M$ is defined by composing the local comparison maps from eq. (4). Although the resulting map is not canonical, its homotopy class (after composition with a marking of $M$) is well defined. See for instance the map in the proof of [BSW22, Proposition 6.1]. A useful feature of real Rel leaves is that all surfaces in the same leaf share the same horizontal foliation type.

Let

$$A = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R})$$

be an upper triangular matrix. Then, the action of the upper triangular group and real Rel is equivariant, that is, whenever $\mathrm{Rel}_t M$ is defined,

$$A \, \mathrm{Rel}_t M = \mathrm{Rel}_{at}(AM),$$

as follows from the linear action of $A$ in period coordinates.

The semidirect product $DU \ltimes \mathbb{R}$ acts on $\widehat{\mathcal{R}}_\infty$ via

$$M \longmapsto (A, t) \cdot M := \mathrm{Rel}_{at}(AM),$$

and this defines a group action. Similarly, the semidirect product $DU \ltimes [0, \infty)$ acts on $\widehat{\mathcal{R}}_{\geq 0}$ via the same formula

$$M \longmapsto (A, t) \cdot M := \mathrm{Rel}_{at}(AM),$$

defining a semigroup action.

3.1. **The real Rel flow on double covers of tori.** The locus $\mathcal{E}$ is invariant under the real Rel flow, as can be seen from its effect on period coordinates and the preservation of area.

Fix a torus $T \in \mathcal{H}^{(1/2)}(0)$ and denote by $\mathcal{E}(T) \subset \mathcal{E}$ the collection of surfaces $M$ such that the quotient $M/\iota_M$ is translation equivalent to $T$.

Strictly speaking, the quotient $M/\iota_M$ has two marked points corresponding to the branch points. However, by forgetting one of these marked points, we may regard $M/\iota_M$ as an element of $\mathcal{H}^{(1/2)}(0)$. From now on, we will omit the specification that $T \in \mathcal{H}^{(1/2)}(0)$ and simply write $T \in \mathcal{H}(0)$, since we may rescale so that $M = T \#_I T$ has area 1.

**Theorem 3.1.** *Let $T \in \mathcal{H}(0)$ be a torus. Then every forward (backward) complete real Rel trajectory in $\mathcal{E}(T)$ is either:*

*(1) Periodic,*

*(2) Dense and equidistributed with respect to the natural probability measure on $\mathcal{E}(T)$.*

**Remark 3.2.** In the situation of item 2, the real Rel flow is uniquely ergodic. Theorem 3.1 follows from the fact that $\mathcal{E}(T)$ can be viewed as a translation surface whose Veech group is a lattice and that the real Rel flow corresponds to the horizontal straight-line flow on it, together with [Sch05, Theorem 1]; see also Theorem 5.3.

**Proposition 3.3.** *Let $T \in \mathcal{H}(0)$ be a torus with no horizontal closed geodesics. The horizontal flow on $M \in \mathcal{E}(T)$ is not minimal if and only if $M = T \#_I T$ where $I \subset T$ is a horizontal segment.*

Let $\alpha \in [0, 1) \setminus \mathbb{Q}$ be irrational, and let $I \subset [0, 1)$ be a subset of the form

$$I = [0, a) \cup [b, 1) \quad \text{or} \quad I = [a, b),$$

where $0 \leq a \leq b \leq 1$ and $0 < |I| < 1$.

Let $X = [0, 1) \times \{-1, 1\}$, where addition on $[0, 1)$ is taken modulo 1 and $\{-1, 1\}$ is endowed with multiplication. Define

$$\mathcal{S}_{I,\alpha} : X \to X, \qquad \mathcal{S}(x, \varepsilon) = (x + \alpha, \, f(x)\varepsilon),$$

where

$$f(x) = \begin{cases} -1 & x \in I, \\ 1 & x \notin I. \end{cases}$$

Accordingly, the map $\mathcal{S}_{[0,1) \setminus I, \alpha}$ is given by

$$(x, \varepsilon) \longmapsto (x + \alpha, \, -f(x)\varepsilon),$$

where $[0,1) \setminus I$ denotes the complement of $I$.

In [Vee69], W. Veech gave conditions on $\alpha$ and the length $|I|$ under which one of the skew products $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is minimal but not uniquely ergodic. This paper is historically significant as it provides one of the first examples of such behavior for interval exchange transformations, although neither the terminology of IETs nor that of translation surfaces appears. The original motivation was to address a different problem, see [Vee69, Theorem 1].

We restate the results of [Vee69] relevant to this paper and indicate where the main ideas of the proof can be found in the original article.

Denote by Leb the Lebesgue measure on $[0,1)$, let $\#$ be the counting measure on the set $\{-1,1\}$. The product $\mathrm{Leb} \otimes \#$ is the natural measure on $[0,1) \times \{-1,1\}$, with

$$\mathrm{Leb} \otimes \# \left([0,1) \times \{-1,1\}\right) = 2.$$

Let $\hat{\imath}$ be the map

$$\hat{\imath} : [0,1) \times \{-1,1\} \longrightarrow [0,1) \times \{-1,1\}, \quad (x,\varepsilon) \longmapsto (x,-\varepsilon).$$

**Theorem 3.4.** *Let $\alpha \in (0,1) \setminus \mathbb{Q}$ and let $I \subset [0,1)$ be as above.*

*(1) If $\alpha$ has unbounded partial quotients, then there exists an uncountable set $K_1$ of measure zero such that if $|I| \in K_1$, only one of the skew products $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is minimal and uniquely ergodic. The other is minimal and admits only two invariant ergodic probability measures $\mu_1$ and $\mu_2$ that are absolutely continuous with respect to* $\mathrm{Leb} \otimes \#$. *The measures $\mu_1$ and $\mu_2$ are exchanged by $\hat{\imath}$, that is*

$$\hat{\imath}^* \mu_1 = \mu_2, \qquad \hat{\imath}^* \mu_2 = \mu_1,$$

*and*

$$\mathrm{Leb} \otimes \# = \mu_1 + \mu_2.$$

*(2) If $\alpha$ has bounded partial quotients and $|I| \notin \{m\alpha + n : m,n \in \mathbb{Z}\}$, then $\mathcal{S}_{I,\alpha}$ and $\mathcal{S}_{[0,1)\setminus I,\alpha}$ are minimal and uniquely ergodic.*

*(3) If $|I| \in \{m\alpha + n : m,n \in \mathbb{Z}\}$, then exactly one of $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is minimal and uniquely ergodic. The other is not minimal and decomposes into two minimal components exchanged by $\hat{\imath}$, each supporting an invariant ergodic probability measure given by the restriction of* $\mathrm{Leb} \otimes \#$.

*References for the proof.* The argument is to reduce the problem to solving a cohomological equation, see [Vee69, Proposition 2]. This is inspired by a classical technique introduced in Furstenberg's work [Hil61]. That is, find a function $g : [0,1) \to \mathbb{C}$ which solves one of the equations

$$g(x+\alpha) = f(x)g(x), \quad \mathrm{Leb}\text{-a.e. } x \tag{6}$$

or

$$g(x+\alpha) = -f(x)g(x), \quad \mathrm{Leb}\text{-a.e. } x. \tag{7}$$

It is proven in [Vee69, Corollary 2], that at least one of the skew products $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is minimal and uniquely ergodic, because eqs. (6) and (7) cannot be solved simultaneously in a non trivial way.

If there is a nontrivial solution for eq. (6) (respectively eq. (7)), we can assume

$$g : [0,1) \to \{-1,1\}.$$

If $g$ is continuous, then $\mathcal{S}_{I,\alpha}$ (respectively $\mathcal{S}_{[0,1)\setminus I,\alpha}$) is not minimal. If $g$ is discontinuous but measurable, then $\mathcal{S}_{I,\alpha}$ (respectively $\mathcal{S}_{[0,1)\setminus I,\alpha}$) is minimal but not uniquely ergodic. The sets

$$A_{-1} = \{(x,-g(x)) : x \in [0,1)\} \quad \text{and} \quad A_1 = \{(x,g(x)) : x \in [0,1)\}$$

are $\mathcal{S}_{I,\alpha}$-invariant (respectively $\mathcal{S}_{[0,1)\setminus I,\alpha}$-invariant), disjoint, have the same non-zero measure and are either open if $g$ is continuous or measurable if $g$ is measurable but not continuous.

The proof of item 1 follows from [Vee69, Theorem 3]. The set $K_1$ depends on $\alpha$ since it is defined using Ostroswski expansions. It turns out that $K_1$ is uncountable if and only if $\alpha$ has unbounded partial quotients. That $K_1$ has measure zero follows from [Vee69, Theorem 2] and the discussion following it. The existence of two invariant ergodic probability measures that are exchanged by $\hat{\imath}$ and whose sum is $\mathrm{Leb} \otimes \#$ follows from the comments preceding [Vee69, Proposition 2].

The proof of item 2 follows from [Vee69, Proposition 7]. Observe that

$$|I| = 1 - |[0,1) \setminus I|$$

then

$$|I|,\ |[0,1)\setminus I| \in \{m\alpha + n : m,n\in\mathbb{Z}\}.$$

The limit

$$h_n^2 = \left(\int_{[0,1)} f(x+(n-1)\alpha)\cdots f(x)\,\mathrm{d}\,\mathrm{Leb}(x)\right)^2 \longrightarrow 1 \quad \text{as} \qquad \|n\alpha\|\to 0$$

is equivalent to solving eq. (6) or eq. (7) by [Vee69, Propositions 4, 5 and 6]. In [Vee69, Proposition 7], it is proved that when $\alpha$ has bounded partial quotients, the limit cannot hold.

Item 3 follows from the first three paragraphs of [Vee69, §4], the argument proves that $h_n^2 \to 1$ as $\|n\alpha\|\to 0$. Again, [Vee69, Propositions 4, 5 and 6] imply that either $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is not ergodic with respect to $\mathrm{Leb}\otimes\#$. The skew product that is not ergodic cannot be minimal, which can be seen using a geometric interpretation that we introduce next and Proposition 3.7. To the best of our knowledge, Veech did not prove this in his paper. See the comment at the end of the proof of [Vee69, Proposition 6]. ☐

3.2. **Geometric interpretation of Veech's skew product.** A geometric interpretation of the skew product $\mathcal{S}$ in the context of translation surfaces appears in [MT02] and [MS91]. We give a similar interpretation to connect Theorem 3.4 with the results of this paper.

Consider the torus $T = \mathbb{R}^2/\Lambda \in \mathcal{H}(0)$ where the lattice is

$$\Lambda = \mathrm{span}\{(0,1),(1,\alpha)\} \tag{8}$$

and $\alpha\in(0,1)$ is irrational. This guarantees that the torus $T$ has no horizontal closed geodesics.

Let $I\subset T$ be a vertical segment of length $|I|$ with one endpoint at the marked point. Let $M = T\#_I T$ be the surface obtained by gluing along the segment $I$. Let $\varsigma\subset M$ be a vertical closed geodesic and define $X = \varsigma\cup\iota(\varsigma)\subset M$. Since the length of $\varsigma$ is 1, the length of $X$ is 2. The first return map of the horizontal flow on $M$ to $X$ is isomorphic to the skew product $\mathcal{S}_{I',\alpha}$ where $I'\subset[0,1)$ with $|I'| = |I|$. This is in fact only a measurable isomorphism with the length form $\mathrm{d}y$ on $X$ and Leb on $[0,1)$. However this causes no difficulty in the sequel since the problem of extending to a homeomorphism occurs on a dense countable subset of $X\subset M$.

A slight modification of the previous construction can be obtained by allowing $I\subset T$ to be any geodesic segment that is not closed. The resulting construction and the associated isomorphism depend only on $\alpha$ and the value of $\mathrm{hol}_T^{(y)}(I)$.

The following lemma allows us to normalize every surface $M'\in\mathcal{E}(T')$ for a given $T'\in\mathcal{H}(0)$. This normalization simplifies many calculations since the horizontal flow on $M'$ can then be understood via Theorem 3.4.

**Lemma 3.5.** *Let $T'$ be a torus with no horizontal closed geodesics. Then there exists an upper triangular matrix $A\in\mathrm{SL}_2(\mathbb{R})$ such that $T = AT'\in\mathcal{H}(0)$ has no horizontal closed geodesics and has a vertical closed geodesic of length* 1.

*Proof.* Suppose that $T' = \mathbb{R}^2/\Lambda'$, where the lattice $\Lambda'$ is of the form

$$\Lambda' = \mathrm{span}\{(a,b),(c,d)\}.$$

Since $T'$ has no horizontal closed geodesics, we must have $d\neq 0$ and $b/d\notin\mathbb{Q}$. Without loss of generality we assume that $\mathrm{area}(T') = ad-bc = 1$.

Consider the matrix

$$A = \begin{pmatrix} d & -c \\ 0 & \frac{1}{d}\end{pmatrix}.$$

The resulting lattice is

$$\begin{aligned}\Lambda = A\Lambda' &= \mathrm{span}\{(ad-bc, b/d),\ (0,1)\}\\ &= \mathrm{span}\{(1,\alpha),\ (0,1)\}\end{aligned}$$

where $\alpha = b/d$.

Hence

$$T = AT' = \mathbb{R}^2/\Lambda,$$

which has a vertical closed geodesic of length 1 and no horizontal closed geodesics. ☐

If $M' = T'\#_{I'}T'$, the matrix $A$ normalizing $T'$ preserves horizontal directions. Hence the horizontal foliations on $M'$ and $M = AM'$ behave in the same way. Consequently, the transverse measures for the horizontal foliations on $M'$ and on $M$ are in one-to-one correspondence with the invariant measures of the skew product $\mathcal{S}_{I,\alpha}$ associated with the normalized surface $M$.

**Remark 3.6.** Item 1 in Theorem 3.4 provides sufficient conditions for the skew product $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ to be minimal and not uniquely ergodic. In [Vee69, Question 2], Veech asked whether these conditions are also necessary. In [CE07, Section 5], the authors answered this question in the affirmative. That is, if one of the skew products $\mathcal{S}_{I,\alpha}$ or $\mathcal{S}_{[0,1)\setminus I,\alpha}$ is minimal and not uniquely ergodic, then necessarily $\alpha$ has unbounded partial quotients and $|I| \in K_1$.

3.3. **Presence of horizontal saddle connections.** We prove Theorem 3.4 (item 3) using the geometric interpretation. We start with the following.

**Proposition 3.7.** *Let $\alpha \in (0,1)$ be irrational. Let $T$ be a torus where the lattice that determines $T$ is as in eq. (8). Let $M \in \mathcal{E}(T)$ be a surface with two horizontal saddle connections. Then, $M$ is obtained by a representation where either*

- *$I$ is a horizontal segment. In this case, the horizontal straight line flow has two minimal components or*
- *$\mathrm{hol}_M^{(y)}(I) \in \{m\alpha + n : m, n \in \mathbb{Z}\}$ and the horizontal straight line flow is minimal and uniquely ergodic.*

*Proof.* This follows from [BSW22, Theorem 7.13 items 2 and 3] in the case $d = 2$.

Let $\alpha \in (0,1)$ be irrational and consider the normalized construction $T\#_I T$ where the lattice that determines $T$ is as in eq. (8) and $I \subset T$ is a vertical geodesic segment with $|I| \in \{m\alpha + n : m, n \in \mathbb{Z}\} \cap (0,1)$. By the choice of the segment $I$, surface $M$ contains two horizontal saddle connections that connect distinct singularities. The union of these horizontal saddle connections forms a closed curve that either separates the surface into two isometric tori or does not separate $M$. In the former case, the horizontal flow on each torus is minimal, hence they are minimal components. Thus the skew product $\mathcal{S}_{I',\alpha}$ with $|I'| = |I|$ is not minimal and therefore not ergodic. In the latter case, the surface $M$ has a polygonal representation as a single torus $T''$ with no horizontal closed geodesics and with two horizontal segments identified along their opposite sides. The horizontal flow on $M$ is therefore minimal and uniquely ergodic since the horizontal flow on $T''$ is uniquely ergodic. Thus the skew product $\mathcal{S}_{I',\alpha}$ is minimal and uniquely ergodic. □

## 4. Tremors

In this section we recall the basics on transverse measure, foliation cocycles and the tremor map.

4.1. **Transverse measures and foliation cocycles.** We recall the basics on transverse measures and foliation cocycles. We refer the reader to [FM12], [CSW25], [MW14] and the references therein for a broader explanation. We will focus on the stratum of area 1 $\mathcal{H}(1,1)$ for simplicity, but these can be extended to the full stratum $\widehat{\mathcal{H}}(1,1)$.

Every translation surface $M$ has a family of singular foliations $\{\Upsilon^\theta\}_{\theta\in[0,\pi)}$ whose leaves are trajectories of the straight-line flow $\Upsilon_t^\theta : M \to M$ in a direction $\theta \in [0,\pi)$. A *transverse measure* $\nu$ for the foliation in direction $\theta \in [0,\pi)$ is a family of measures $\{\nu_\gamma\}$ supported on curves $\gamma$ transverse to the foliation $\Upsilon^\theta$, satisfying invariance under isotopies that keep the endpoints of $\gamma$ on the leaves of the foliation, as well as the restriction properties of measures. We will only focus on *non-atomic* transverse measures, that is, the measures $\nu_\gamma$ are finite, regular Borel measures and no $\nu_\gamma$ has atoms. When we say transverse measure, we mean: *transverse measure for the horizontal flow* ($\theta = 0$). When $\theta = 0$, the canonical transverse measure is determined by the real 1-form $\mathrm{d}y$, and when $\theta = \frac{\pi}{2}$, the canonical transverse measure for the vertical foliation is $\mathrm{d}x$. There is a correspondence between invariant measures for the flow $\Upsilon_t^\theta$ and transverse measures for the foliation $\Upsilon^\theta$. This is due to Katok; see [CSW25, Proposition 2.3] and the references therein. Essentially, if $\mu$ is a $\Upsilon_t^\theta$-invariant non-atomic finite measure on $M$, then using charts we can decompose $\mu$ as a product of a transverse measure $\nu$ for $\Upsilon^\theta$ and Lebesgue measure along the leaves using the translation structure of $M$:

$$\mu(A) = \nu(u)\,\ell(h), \tag{9}$$

where $A \subset M$ is a sufficiently small rectangle with side $u$ orthogonal to the direction $\theta$ and side $h$ contained in a leaf of the foliation $\Upsilon^\theta$, and $\ell(h)$ is the Euclidean length of the segment $h$. When $\mu = \mathrm{Leb}$ and $\theta = 0$, we can write $\mu = \mathrm{d}x \wedge \mathrm{d}y$. We have the following bijection:

$$\left\{\text{non-atomic } \Upsilon_t^\theta\text{-invariant measures on } M\right\} \longleftrightarrow \left\{\text{non-atomic transverse measures for } \Upsilon^\theta\right\}.$$

The previous discussion extends to signed invariant measures and signed transverse measures via the Hahn decomposition:

$$\mu = \mu^+ - \mu^- \quad \Longleftrightarrow \quad \nu = \nu^+ - \nu^-.$$

Given a transverse measure $\nu$ for $\Upsilon^\theta$, we define a *foliation cocycle*

$$\beta_\nu \in H^1(M, \Sigma; \mathbb{R})$$

by

$$\beta_\nu([\gamma]) := \int_\gamma \mathrm{d}\nu = \nu(\gamma),$$

for any $[\gamma] \in H_1(M, \Sigma; \mathbb{Z})$, where $\gamma$ is an oriented representative whose angle measured counter-clockwise with respect to the direction $\theta$ lies in $(0, \pi)$. If the angle lies in $(-\pi, 0)$, we set

$$\beta_\nu([\gamma]) := -\int_\gamma \mathrm{d}\nu = - = \nu(\gamma).$$

Using the identification

$$H^1(M, \Sigma; \mathbb{R}^2) \simeq H^1(M, \Sigma; \mathbb{R}) \oplus H^1(M, \Sigma; \mathbb{R}),$$

we defined the map

$$\nu \longmapsto \beta_\nu \in H^1(M, \Sigma; \mathbb{R}) \oplus 0.$$

We regard foliation cocycles, as cocycles pointing to the "horizontal" period coordinates, this will allows us to obtain "nice" interactions with the upper triangular matrices of $\mathrm{SL}(2, \mathbb{R})$ and the real Rel flow. Let $M \in \mathcal{H}(1,1)$ be a translation surface. For simplicity, we write $\Upsilon := \Upsilon^0$ its horizontal foliation and denote

$$C_M^+ := \{\beta_\nu : \nu \text{ is a transverse measure for } \Upsilon\}$$

and

$$\mathcal{T}_M := \{\beta_1 - \beta_2 : \beta_1, \beta_2 \in C_M^+\},$$

the collections of foliation cocycles and *signed foliation cocycles* for $M$, respectively. We define the *signed mass* of $\beta_\nu \in \mathcal{T}_M$ (for $\nu$ signed) by the cup product

$$L_M(\beta_\nu) := \int_M \mathrm{d}\mu = \int_M \mathrm{d}x \wedge \mathrm{d}\nu,$$

where $\mu$ is the measure the identification above and eq. (9). The foliation cocycles in the collection

$$\mathcal{T}_M^{(0)} := \{\beta \in \mathcal{T}_M : L_M(\beta) = 0\}$$

are called *balanced foliation cocycles*.

The map $\nu \mapsto \beta_\nu$ is injective if $M$ has no horizontal closed geodesics. Closed curves give rise to atomic transverse measures, atomic invariant measures and atomic foliation cocycles. This allows different transverse measures to give rise to the same foliation cocycle. Again, we do not consider atomic transverse measures nor atomic foliation cocycles.

The total variation of a foliation cocycle $\beta \in \mathcal{T}_M$ is defined by

$$|L|_M(\beta) := L_M(\beta^+) + L_M(\beta^-),$$

where $\beta = \beta^+ - \beta^-$ is obtained by the Hahn decomposition of the signed transverse measure that determined $\beta$. The transverse measure $\mathrm{d}y$ gives rise to the canonical foliation cocycle $\mathrm{hol}_M^{(x)}$ and

$$|L|_M(\mathrm{hol}_M^{(x)}) \;=\; \mathrm{Leb}(M) =\; 1.$$

The signed conventions imply that for all $\beta \in C_M^+$, then $\beta^- = 0$ and $|L|_M(\beta) = L_M(\beta) \geq 0$.

If $h : M \to M$ is a translation equivalence, then for all $\beta \in \mathcal{T}_M$,

$$L_M(h_*\beta) = \int_M \mathrm{d}x \wedge \mathrm{d}(h_*\nu) = \int_M \mathrm{d}x \wedge \mathrm{d}\nu = L_M(\beta),$$

since $\mathrm{d}x$ is invariant under $h$. Likewise $|L|_M(h_*\beta) = |L|_M(\beta)$. This is important since the tangent space at $M$ is of the form $H^1(S, \Sigma; \mathbb{R}^2)/\operatorname{Stab}(M, f)$ and will be use implicitly in our calculations for orbit closures of Section 6.

Given a marking $(M,f)\in\mathcal{H}_m$, the pullbacks $f^*(\mathcal{T}_M)$ and $f^*(C_M^+)$ define subsets of $H^1(S,\Sigma;\mathbb{R})$. These do not depend on the representative of $(M,f)$. We denote them by $\mathcal{T}_{(M,f)}$ and $C^+_{(M,f)}$, respectively. For $\tilde{\beta}\in\mathcal{T}_{(M,f)}$, we define the signed mass

$$L_{(M,f)}(\tilde{\beta}) := L_M(f_*\tilde{\beta}).$$

This is well-defined since all markings are isotopic. For $(M,f)\in\mathcal{H}_m(1,1)$, we have

$$f^*(\mathcal{T}_M^{(0)}) = \{\tilde{\beta}\in\mathcal{T}_{(M,f)} : L_{(M,f)}(\tilde{\beta})=0\} =: \mathcal{T}^{(0)}_{(M,f)}.$$

Similarly, the total variation for $\tilde{\beta}\in\mathcal{T}_{(M,f)}$ is defined by

$$|L|_{(M,f)}(\tilde{\beta}) := L_M((f_*\tilde{\beta})^+) + L_M((f_*\tilde{\beta})^-).$$

By linearity of the cup product, the maps

$$\{(M,\beta) : M\in\mathcal{H},\ \beta\in\mathcal{T}_M\}\longrightarrow\mathbb{R},\quad (M,\beta)\longmapsto L_M(\beta),$$

and

$$\{((M,f),\widetilde{\beta}) : (M,f)\in\mathcal{H}_m,\ \widetilde{\beta}\in\mathcal{T}_{(M,f)}\}\longrightarrow\mathbb{R},\quad ((M,f),\tilde{\beta})\longmapsto L_{(M,f)}(\tilde{\beta})$$

are continuous. However, the total variation is only semicontinuous: if $(M_i,\beta_i)\to(M_\infty,\beta_\infty)$, then

$$|L|_{M_\infty}(\beta_\infty)\leq\liminf_{i\to\infty}|L|_{M_i}(\beta_i).$$

See [CSW25, Corollary 4.4]. This is an inconvenience for carrying calculations in the limit. To solve this issue, we do arguments in the marked stratum $\mathcal{H}_m$ and use a norm $\|\cdot\|$ on the tangent space $H^1(S,\Sigma;\mathbb{R}^2)$ that varies continuously (and defines a distance in $\mathcal{H}$) with respect to the topology of the period coordinates. This is the case of the proof of Proposition 5.1 or Lemma 6.7.

If $M\in\mathcal{H}(1,1)$ has no horizontal closed geodesics, then the flow $\Upsilon_t$ has at most two ergodic invariant probability measures. Hence $C_M^+$ is a convex cone with at most two extremal rays. We denote the set of extremal rays by $C_M^{+,\mathrm{erg}}$.

Let $a\geq 0$. Then, the sets

$$\{(M,\beta) : M\in\mathcal{H}(1,1),\ \beta\in C_M^+,\ L_M(\beta)=a\}\quad\text{and}\quad\{(M,\beta) : M\in\mathcal{H}(1,1),\ \beta\in\mathcal{T}_M^{(0)},\ |L|_M(\beta)\leq a\}$$

are closed. Their lifts to the marked stratum are closed as well.

4.2. **Tremor maps.** Given $(M,f)\in\widehat{\mathcal{H}}_m(1,1)$ and $\widetilde{\beta}\in\mathcal{T}_{(M,f)}\subset H^1(S,\Sigma;\mathbb{R})$, let

$$\Psi_t^{(M,f,\widetilde{\beta})} : D\longrightarrow\widehat{\mathcal{H}}_m(1,1)$$

be a solution of the ODE

$$\Psi_0^{(M,f,\widetilde{\beta})}=(M,f),\quad \frac{d}{dt}\Psi_t^{(M,f,\widetilde{\beta})}=\widetilde{\beta}\oplus 0.$$

The solution is unique and has a maximal domain of definition $D:=D((M,f),\widetilde{\beta})$. It was shown in [CSW25] that for all $(M,f)\in\widehat{\mathcal{H}}_m(1,1)$ and all $\widetilde{\beta}\in\mathcal{T}_{(M,f)}$, the domain of definition is of the form

$$D((M,f),\widetilde{\beta})=\mathbb{R}.$$

We define the tremor map

$$\mathrm{trem}_{\widetilde{\beta}}(M,f) := \Psi_1^{(M,f,\widetilde{\beta})}.$$

By uniqueness of solutions, there is an $\mathbb{R}$-action: for every $s\in\mathbb{R}$,

$$\mathrm{trem}_{s\widetilde{\beta}}(M,f)=\Psi_s^{(M,f,\widetilde{\beta})}.$$

In period coordinates, for every $\gamma\in H^1(S,\Sigma;\mathbb{R}^2)$,

$$\mathrm{hol}^{(x)}_{\mathrm{trem}_{\widetilde{\beta}}(M,f)}(\gamma)=\mathrm{hol}^{(x)}_{(M,f)}(\gamma)+\widetilde{\beta}(\gamma)\quad\text{and}\quad\mathrm{hol}^{(y)}_{\mathrm{trem}_{\widetilde{\beta}}(M,f)}(\gamma)=\mathrm{hol}^{(y)}_{(M,f)}(\gamma).$$

Moreover, area of the marked surfaces $\Psi_t^{(M,f,\widetilde{\beta})}$ is constant in $t$, since the the tremor map if a deformation along the horizontal foliation of the underlying translation surface. Additionally, for

$$\widetilde{\beta}=s\,f^*\,\mathrm{hol}_M^{(y)},$$

the previous discussion implies that for $(M, f) \in \mathcal{H}_m$ (area 1),

$$\operatorname{trem}_{s\, f^* \operatorname{hol}_M^{(y)}}(M, f) = (u_s M, \phi_{u_s} \circ f),$$

where $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ and $\phi_{u_s}$ is the corresponding affine comparison map.

The group $\operatorname{Mod}(S, \Sigma)$ acts equivariantly on both factors of

$$\widehat{\mathcal{H}}_m(1,1) \times H^1(S, \Sigma; \mathbb{R}^2)$$

with respect to the developing map and preserves area. Hence, for $M \in \mathcal{H}(1,1)$ and $\beta \in \mathcal{T}_M$, we denote by $\operatorname{trem}_\beta M \in \mathcal{H}(1,1)$, the projection of $\operatorname{trem}_{f^*\beta}(M, f) \in \mathcal{H}_m(1,1)$.

By standard properties of ODEs, the tremor map

$$\mathfrak{T}_m : \{((M,f), \widetilde{\beta}) : (M,f) \in \mathcal{H}_m(1,1),\ \widetilde{\beta} \in \mathcal{T}_{(M,f)}\} \longrightarrow \mathcal{H}_m(1,1), \quad ((M,f), \widetilde{\beta}) \longmapsto \operatorname{trem}_{\widetilde{\beta}}(M,f)$$

is continuous where defined. Similarly, the map

$$\mathfrak{T} : \{(M, \beta) : M \in \mathcal{H}(1,1),\ \beta \in \mathcal{T}_M\} \longrightarrow \mathcal{H}(1,1), \quad (M, \beta) \longmapsto \operatorname{trem}_\beta M$$

is continuous where defined.

In the marked stratum $\mathcal{H}_m(1,1)$, we have a group action law. Suppose that

$$(M,f) \in \mathcal{H}_m(1,1), \qquad \widetilde{\beta}_1 \in \mathcal{T}_{(M,f)} \quad \text{and} \quad \widetilde{\beta}_2 \in \mathcal{T}_{\operatorname{trem}_{\widetilde{\beta}_1}(M,f)}.$$

In [CSW25, Proposition 4.14] and their discussion before it, they show that the $\mathcal{T}_{(M,f)}$ and $\mathcal{T}_{\operatorname{trem}_{\widetilde{\beta}_1}(M,f)}$ have a natural identification. Thus, by abuse of notation there exists $\widetilde{\beta}_2 \in \mathcal{T}_{(M,f)}$ such that

$$\operatorname{trem}_{\widetilde{\beta}_1 + \widetilde{\beta}_2}(M,f) = \operatorname{trem}_{\widetilde{\beta}_2}\Big(\operatorname{trem}_{\widetilde{\beta}_1}(M,f)\Big).$$

By definition of tremors and equivariance of the $\operatorname{Mod}(S, \Sigma)$-action, we obtain

$$\operatorname{trem}_{\beta_2} \operatorname{trem}_{\beta_1} M = \operatorname{trem}_{\beta_1 + \beta_2} M,$$

for $\beta_2 \in \mathcal{T}_{\operatorname{trem}_{\beta_1} M}$, and the corresponding cocycles $\beta_1, \beta_2 \in \mathcal{T}_M$. Moreover, these identifications preserve signed mass and total variation:

$$L_M(\beta_2) = L_{\operatorname{trem}_{\beta_1} M}(\beta_2), \quad \text{and} \quad |L|_M(\beta_2) = |L|_{\operatorname{trem}_{\beta_1} M}(\beta_2).$$

4.3. **Interaction with upper triangular matrices and real Rel.** For

$$A = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \in DU,$$

we obtain identifications of $\mathcal{T}_M$ and $\mathcal{T}_{AM}$, since the action of matrices in $DU$ is associated to affine comparison maps that preserve the horizontal foliation. Similarly, the real Rel flow comes equipped with affine map that preserve the horizontal foliation. Denote these maps by

$$\phi_A : M \longrightarrow AM \quad \text{and} \quad \phi_t : M \longrightarrow \operatorname{Rel}_t M$$

and the corresponding identification of signed foliation cocycles

$$\phi_A^* : \mathcal{T}_M \longrightarrow \mathcal{T}_{AM} \quad \text{and} \quad \phi_t^* : \mathcal{T}_M \longrightarrow \mathcal{T}_{\operatorname{Rel}_t M}.$$

These identification imply for $\beta \in \mathcal{T}_M$ the relations

$$\begin{aligned} L_M(\beta) &= a\, L_{AM}(\phi_A^*\beta), & |L|_M(\beta) &= a\, |L|_{AM}(\phi_A^*\beta), \\ L_M(\beta) &= L_{\operatorname{Rel}_t M}(\phi_t^*\beta) \quad \text{and} & |L|_M(\beta) &= |L|_{\operatorname{Rel}_t M}(\phi_t^*\beta). \end{aligned}$$

For simplicity, we often write $\beta = \phi_A^*\beta$ or $\beta = \phi_t^*\beta$, although these cocycles belong to different surfaces. Suppose that $\operatorname{Rel}_t M$ is defined, $\beta \in \mathcal{T}_M$, and $A \in DU$. Then $\operatorname{Rel}_t \operatorname{trem}_\beta M$ is defined, and

$$A \operatorname{Rel}_t \operatorname{trem}_\beta M = \operatorname{trem}_{a\beta} \operatorname{Rel}_{at} AM.$$

A useful property for proving statements using period coordinates is the tremor comparison homeomorphism established in [CSW25, Proposition 5.13].

**Theorem 4.1** (Tremor comparison homeomorphism (TCH))**.** *Let $(M,f)\in\mathcal{H}_m(1,1)$ be a marking and let $\beta_\nu = f_*\widetilde{\beta}\in\mathcal{T}_M$ be a foliation cocycle corresponding to a transverse measure $\nu$. Denote by*

$$(M_t,f_t) := \mathrm{trem}_{\widetilde{\beta}}(M,f)\in\mathcal{H}_m(1,1).$$

*If $f_t : S\longrightarrow M_t$ is a map representing $(M_t,f_t)$, then there exists a unique homeomorphism (the TCH)*

$$h_t : M\longrightarrow M_t$$

*which is isotopic to $f_t\circ f^{-1}$, preserves the horizontal foliation and satisfies*

$$\mathrm{hol}^{(x)}_{M_t}(h_t(\gamma)) = \mathrm{hol}^{(x)}_M(\gamma) + t\int_\gamma \mathrm{d}\nu \quad \textit{and} \quad \mathrm{hol}^{(y)}_{M_t}(h_t(\gamma)) = \mathrm{hol}^{(y)}_M(\gamma),$$

*for every piecewise smooth path $\gamma$ connecting two points in $M$.*

The following lemma is a special case for foliation cocycles of surfaces belonging to the orbifold locus $\mathcal{E}\subset\mathcal{H}(1,1)$

**Lemma 4.2.** *Let $T\in\mathcal{H}(0)$ have no horizontal closed geodesics and let $M\in\mathcal{E}(T)$ have non-ergodic horizontal flow. If $\beta\in\mathcal{T}^{(0)}_M$ with $|L|_M(\beta)=a>0$, then*

$$\beta=\beta_1-\beta_2$$

*for some distinct $\beta_1,\beta_2\in C^{+,erg}_M$ with $L_M(\beta_1)=L_M(\beta_2)=a/2$.*

*If $\beta'\in\mathcal{T}_M$ satisfies $L_M(\beta')=c\in\mathbb{R}$, then $\beta'-c\,\mathrm{hol}^{(y)}_M\in\mathcal{T}^{(0)}_M$ and*

$$|L_M(\beta'-c\,\mathrm{hol}^{(y)}_M)|\le|L_M(\beta')|.$$

*Proof.* Write $\beta=\beta^+-\beta^-$ with $\beta^\pm\in C^+_M$. Since the horizontal flow on $M$ has exactly two ergodic invariant measures, $C^+_M$ is generated by $\beta_1,\beta_2\in C^{+,erg}_M$, which we normalize so that $L_M(\beta_i)=a/2$. Thus $\beta^\pm$ are nonnegative combinations of $\beta_1,\beta_2$. We can write

$$\beta=e_1\beta_1+e_2\beta_2.$$

The condition $L_M(\beta)=0$ implies $e_1=-e_2$. After relabeling, we may assume $e_2<0$ and write $e=e_1>0$, so that $\beta=e(\beta_1-\beta_2)$. Since $|L|_M(\beta)=a$ and $L_M(\beta_i)=a/2$, we obtain $e=1$.

We prove the last statement. Without loss of generality, assume that $\mathrm{Leb}(M)=1$. Since $\mathcal{T}_M$ is generated by two positive foliation cocycles, we may write

$$\beta'=b_1\beta_1'+b_2\beta_2' \quad\text{and}\quad \mathrm{hol}^{(y)}_M=\frac{1}{2}\beta_1'+\frac{1}{2}\beta_2',$$

where $\beta_i'\in C^{+,erg}_M$ for $i=1,2$. For convenience, we normalize so that

$$L_M(\beta_i')=1.$$

Write

$$c=L_M(\beta')=b_1+b_2.$$

By linearity of $L_M$, it follows that

$$L_M(\beta'-c\ \mathrm{hol}^{(y)}_M)=0\quad\text{hence}\quad \beta'-c\ \mathrm{hol}^{(y)}_M\in\mathcal{T}^{(0)}_M.$$

Moreover,

$$\beta'-c\ \mathrm{hol}^{(y)}_M=\frac{b_1-b_2}{2}\beta_1'-\frac{b_1-b_2}{2}\beta_2',$$

so

$$|L|_M(\beta'-c\ \mathrm{hol}^{(y)}_M)=|L|_M\left(\frac{b_1-b_2}{2}\beta_1'-\frac{b_1-b_2}{2}\beta_2'\right)=|b_1-b_2|.$$

Therefore,

$$|L|_M(\beta'-c\ \mathrm{hol}^{(y)}_M)=|b_1-b_2|\le|b_1|+|b_2|=|L|_M(\beta').$$

$\square$

4.4. **Renormalization.** The result below follows [CSW25, Proposition 3.5], but yields a stronger statement: there exists a universal sequence $t_j$ valid for all $M \in \mathcal{E}(T)$ whose horizontal flow is not uniquely ergodic. The proof adapts their argument essentially verbatim, with additional explicit calculations.

**Lemma 4.3.** *Let $T \in \mathcal{H}(0)$ be a torus with no horizontal closed geodesics. Then there exist a sequence $t_j \to \infty$ and a compact set $K \subset \mathcal{H}(0)$ such that $g_{-t_j}T \in K$ for all $j$.*

*Let $\{t_{j_k}\}$ be a subsequence along which $g_{-t_{j_k}}T$ converges. For every $M \in \mathcal{E}(T)$ whose horizontal flow is not ergodic and every $\varepsilon > 0$, there exists $\widetilde{t} > 0$ such that for all $t_{j_k} \geq \widetilde{t}$,*

$$g_{-t_{j_k}}M = (g_{-t_{j_k}}T)\#_{J_{j_k}}(g_{-t_{j_k}}T),$$

*with*

$$|\operatorname{hol}^{(x)}_{g_{-t_{j_k}}T}(J_{j_k})| + |\operatorname{hol}^{(y)}_{g_{-t_{j_k}}T}(J_{j_k})| < \varepsilon.$$

*If the horizontal flow on $M$ is minimal, then there exist segments $I_{j_k} \subset T$ such that*

$$M = T\#_{I_{j_k}}T, \qquad |\operatorname{hol}^{(x)}_T(I_{j_k})| + |\operatorname{hol}^{(y)}_T(I_{j_k})| \to \infty \quad \text{as } t_{j_k} \to \infty,$$

*where*

$$\operatorname{hol}^{(x)}_T(I_{j_k}) = e^{t_{j_k}} \operatorname{hol}^{(x)}_{g_{-t_{j_k}}T}(J_{j_k}), \qquad \operatorname{hol}^{(y)}_T(I_{j_k}) = e^{-t_{j_k}} \operatorname{hol}^{(y)}_{g_{-t_{j_k}}T}(J_{j_k}).$$

**Remark 4.4.** We emphasize that the sequence $t_j$ depends only on $T$ and is independent of $M$. Only $\widetilde{t}$ depends on $M$ and $\varepsilon$. For all sufficiently large $j$, we can represent $g_{-t_j}M$ as a connected sum of two tori glued along a geodesic segment that is arbitrarily short.

*Proof.* Without loss of generality, we may assume that $M$ and $T$ are normalized as in Section 3.2 and Lemma 3.5. Let $\alpha \in (0,1)$ be the irrational defined in eq. (8). Let $\alpha = [a_1, a_2, \dots]$ be its continued fraction expansion and let $p_j/q_j$ be its best approximants. Let $T_j = g_{-t_j}T$ be the sequence of tori with $t_j = \log q_j$ and

$$g_{-t_j} = \begin{pmatrix} \frac{1}{q_j} & 0 \\ 0 & q_j \end{pmatrix}.$$

The lattice defining $T_j = \mathbb{R}^2/\Lambda_j$ is

$$\Lambda_j = \operatorname{span}\{(1, (-1)^j q_j \|q_j\alpha\|), (q_{j-1}/q_j, (-1)^{j-1} q_j \|q_{j-1}\alpha\|)\}, \tag{10}$$

where we used the identity $q_j\alpha - p_j = (-1)^j \|q_j\alpha\|$. By the theory of continued fractions, we have the uniform lower bound

$$\frac{1}{2} < \frac{a_j}{a_j+1} < \frac{q_j}{q_j+q_{j-1}} < q_j \|q_{j-1}\alpha\| < 1.$$

Hence, for all $j$, every closed geodesic in $T_j$ has length greater than $\frac{1}{2}$. Let $K \subset \mathcal{H}(0)$ be the set of tori for which the shortest closed geodesic has length at least $\frac{1}{2}$. Then $K$ is compact.

We split the remainder of the proof into two cases.

**Case 1** ($M$ is not minimal)**.** By Proposition 3.3, there exists a polygonal representation $M = T\#_I T$ with a horizontal segment $I \subset T$. The affine map corresponding to $g_{-t_j}$ implies that

$$M_j = g_{-t_j}M = T_j\#_{J_j}T_j,$$

with

$$\operatorname{hol}_{T_j}(J_j) = \left(\frac{x}{q_j}, 0\right),$$

where $\operatorname{hol}_T(I) = (x, 0)$. This proves the claim in this case.

**Case 2** ($M$ is minimal)**.** We argue as in [CSW25, Proposition 3.5]. Let $\mathcal{M}_g$ be the moduli space of Riemann surfaces of genus $g$ and let $\overline{\mathcal{M}}_g$ be its Deligne-Mumford compactification. Fixing any subsequence of $t_j$ and passing through a further subsequence (which we still denote by $t_j$), we may assume that the projections of $M_j = g_{-t_j}M$ converge to a stable curve $Y \in \overline{\mathcal{M}}_2$ and that $T_j$ converges to a non-degenerate torus $Y' \in \mathcal{M}_1$, since the compact set $K$ projects to a compact set $K' \subset \mathcal{M}_1$.

Since $Y'$ is a torus, the limit $Y$ has positive area. By [McM13a, Theorem 1.4], if $Y^*$ denotes the smooth part of $Y$, then $Y^*$ has two connected components because the horizontal flow on $M$ is not uniquely ergodic. Thus, $Y$ is the connected sum of two copies of $Y'$ at a point.

The previous argument implies that we can write

$$M_j = T_j \#_{J_j} T_j,$$

where $J_j \subset T_j$ is a segment with $|J_j| \to 0$ as $t_j \to \infty$. Hence, there exists $\widetilde{t} > 0$ such that

$$|\operatorname{hol}^{(x)}_{T_j}(J_j)| + |\operatorname{hol}^{(y)}_{T_j}(J_j)| < \varepsilon \quad \text{for all} \quad t_j > \widetilde{t}.$$

By Proposition 3.3, we have that $\operatorname{hol}^{(y)}_{T_j}(J_j) \neq 0$. The affine map induced by $g_{t_j}$ maps the segment $J_j$ to a segment $I_j \subset T$. This implies that

$$M = T \#_{I_j} T.$$

Denote $x_j = \operatorname{hol}^{(x)}_{T}(I_j)$ and $y_j = \operatorname{hol}^{(y)}_{T}(I_j)$. Then

$$\begin{aligned} \varepsilon &> |\operatorname{hol}^{(x)}_{T_j}(J_j)| + |\operatorname{hol}^{(y)}_{T_j}(J_j)| \\ &= \frac{|x_j|}{q_j} + |y_j| q_j. \end{aligned}$$

We claim that

$$|x_j| + |y_j| = |\operatorname{hol}^{(x)}_{T}(I_j)| + |\operatorname{hol}^{(y)}_{T}(I_j)| \longrightarrow \infty \quad \text{as} \quad t_j \to \infty.$$

By contradiction, suppose that there exists $C > 0$ such that $|x_j| + |y_j| \leq C$ for all $j$. The set of holonomies of saddle connections on $M$ is discrete. Therefore, the sequence $\{I_j\}$ is eventually constant after passing to a subsequence.

Dividing by $q_j$, we obtain that $|y_j| < \varepsilon / q_j \to 0$ as $j \to \infty$. Hence the sequence $\{I_j\}$ is eventually constant and horizontal. This contradicts the minimality of $M$ by Proposition 3.3.

□

We will use the following result from [CSW25] in a form adapted to our setting.

**Theorem 4.5** ([CSW25, Proposition 3.5])**.** *Let $T \in \mathcal{H}(0)$ be a torus with no horizontal closed geodesics. Suppose that the horizontal flow on $M = T \#_I T \in \mathcal{E}(T)$ is minimal but not uniquely ergodic. Then the following holds.*

*(1) There exists a sequence of segments $I_j \subset T$ such that $M = T \#_{I_j} T$,*

$$|x_j| = |\operatorname{hol}^{(x)}_{T}(I_j)| \to \infty \quad \textit{and} \quad y_j = \operatorname{hol}^{(y)}_{T}(I_j) \longrightarrow 0 \quad \textit{as} \quad j \to \infty.$$

*(2) If $\mu$ is an invariant ergodic probability measure on $M$ for the horizontal flow, there exists a sequence of connected components*

$$A_j \subset M \setminus I_j \cup \iota(I_j)$$

*such that the normalized restriction $\mu_j$ of* Leb *to $A_j$ converges to $\mu$ in the weak-$*$ topology of probability measures on $M$.*

*(3) If $\nu$ and $\nu_j$ are the transverse measures corresponding to the measures $\mu$ and $\mu_j$. Letting $\beta_{\nu_j}$, $\beta_\nu$ be the corresponding foliation cocycles in $H_1(M, \Sigma; \mathbb{R})$, then*

$$\beta_{\nu_j} \longrightarrow \beta_\nu.$$

Item 1 is proved as Claim 2 in the proof of Lemma 4.3. Items 2 and 3 follow verbatim from [CSW25, Proposition 3.5].

## 5. Equidistribution of real Rel trajectories

In this section, we study invariant measures for the real Rel flow and prove equidistribution results. We begin by introducing a family of surfaces.

5.1. **Real Rel trajectories on modular fibers.** Let $M = (M, x_1, x_2) \in \mathcal{H}(1,1)$ be a translation surface with marked singularities $x_1$ and $x_2$. We say that $M$ is a *translation d-fold branched cover* (for short, branched cover) of a torus

$$T = (T, z_1, z_2) \in \mathcal{H}(0,0)$$

with two distinct marked points $z_1 \neq z_2$ if there exists a map

$$\pi = \pi_{M,T} : M \to T \tag{11}$$

that is a translation in local coordinates, maps the singularities of $M$ to the marked points of $T$ and whose restriction

$$\pi\big|_{M\setminus R} : M \setminus R \longrightarrow T \setminus \{z_1, z_2\}$$

is a $d$-fold covering map. We write $R = \pi^{-1}(\pi(\{x_1, x_2\}))$ for the ramification points.

Since tori are homogeneous surfaces, we may always assume that the first marked point is $z_1 = 0$.

Consider the map

$$F : \mathcal{H}(0,0) \longrightarrow \mathcal{H}(0), \qquad (T, 0, z) \longmapsto (T, 0), \tag{12}$$

that forgets the second marked point. This is a continuous map since it can be thought as the forgetful map of the second coordinate of the isomorphism

$$\mathcal{H}(0,0) \simeq \big((\mathrm{SL}(2,\mathbb{R}) \ltimes \mathbb{R}^2)/(\mathrm{SL}(2,\mathbb{Z}) \ltimes \mathbb{Z}^2)\big) \setminus ((\mathrm{SL}(2,\mathbb{R})/\mathrm{SL}(2,\mathbb{Z})) \ltimes \{0\}).$$

Denote by $\mathcal{F}_d(T)$ the collection of translation surfaces in $\mathcal{H}(1,1)$ which are $d$-fold branched covers of a torus in

$$F^{-1}((T,0)) \subset \mathcal{H}(0,0).$$

The set $\mathcal{F}_d(T)$ is called a *modular fiber*.

Define also

$$\Pi_T : \mathcal{F}_d(T) \longrightarrow T \setminus \{0\}, \qquad (M, x_1, x_2) \longmapsto \pi(x_2),$$

which assigns $M \in \mathcal{F}_d(T)$ to the second point of the base with the branched cover in (11).

The notation above is inspired by [Sch05]. We replace $\mathbb{T} = \mathbb{R}^2/\mathbb{Z}^2$ by a general flat torus $T \in \mathcal{H}(0,0)$. Instead of writing $\mathcal{F}_d(T)(1,1) \subset \mathcal{H}(1,1)$, we abbreviate the notation to $\mathcal{F}_d(T)$. See also [EMS03, § 3].

The map $\Pi_T$ is a covering map of finite degree.

$$z \in T \setminus \{0\}.$$

The fiber $\Pi_T^{-1}(z)$ for $z \in T$ consists of translation surfaces $M \in \mathcal{H}(1,1)$ equipped with a $d$-fold branched cover of $(T, 0, z)$. By covering theory, there are only finitely many such coverings up to topological type and after pulling back the 1-form $\mathrm{d}z$ on $T$, there are still only finitely many possibilities. The exact degree of $\Pi_T$ was calculated in [EMS03, Corollary 3.3].

In [EMS03], the set $\mathcal{M}_d(1,1) \subset \mathcal{H}(1,1)$ is the union

$$\mathcal{M}_d(1,1) = \bigcup_{T \in \mathcal{H}^{(1/d)}(0)} \mathcal{F}_d(T),$$

where $\mathcal{H}^{(1/d)}(0)$ is the collection of tori of area $1/d$. In [EMS03, Lemma 2.1], it was prove that $\mathcal{M}_d(1,1)$ is closed and $\mathrm{SL}(2,\mathbb{R})$-invariant. To simplify notation, denote by

$$\pi_T : \mathcal{F}_d(T) \longrightarrow F^{-1}((T,0)) \subset \mathcal{H}(0,0), \qquad M \longmapsto \pi_{M,T}(M) = (T, 0, z), \tag{13}$$

the map that sends each $d$-fold covering surface $M \in \mathcal{F}_d(T)$ to the base it covers, together with its marked points. This is a covering map and can be seen as the restriction $p|_{\mathcal{F}_d(T)}$ to the fiber $\mathcal{F}_d(T) = \mathcal{M}_d(1,1) \cap F^{-1}((T,0))$ of the covering map

$$p : \mathcal{M}_d(1,1) \longrightarrow \mathcal{H}(0,0) \tag{14}$$

of [EMS03, Lemma 2.2].

The fact that $\mathcal{F}_d(T)$ is connected follows from [Sch05, Theorem 2]. The collection $\mathcal{F}_d(T)$ is closed as a subset of $\mathcal{H}(1,1)$ and by adding finitely many nodal surfaces and surfaces in $\mathcal{H}(2)$, it admits a compactification. The number of points in this compactification is proved in [Sch05, eq. (4)]. For this reason, we will refer to $\mathcal{F}_d(T) \subset \mathcal{H}(1,1)$ as a translation surface, even though the ambient space $\mathcal{H}(1,1)$ does not contain the singularities required in the formal definition of a translation surface.

When $d = 2$, we write

$$\mathcal{E}(T) := \mathcal{F}_2(T)$$

following the notation of [CSW25] and [BSW22] where $\mathcal{E} = \mathcal{E}_4 \subset \mathcal{H}(1,1)$ is the locus $\mathcal{M}_2(1,1)$. The families $\mathcal{M}_d(1,1)$ are examples of rank-one loci introduced by [Wri15] and related to the eigenform loci of real multiplication with square discriminant $D = d^2$. The later were introduced in [McM07a], see [BSW22, § 7] for more details.

We are now able to state the following equidistribution result.

**Proposition 5.1.** *Let $T \in \mathcal{H}(0)$ be a torus with no horizontal closed geodesics. Let $M \in \mathcal{F}_d(T)$ be a surface, such that $\mathrm{Rel}_t\, M$ is defined for all $t \geq 0$. For any $\beta \in C_M^+$ the surface $\mathrm{trem}_\beta\, M$ is a generic point for $\mu_{u_{s_0}T}$, where $s_0 = L_M(\beta)$.*

In [CSW25, Proposition 8.1], it was proved that $\mathrm{trem}_\beta\, M \notin \mathcal{M}_d(1,1)$ for every $M \in \mathcal{F}_d(T)$ and every foliation cocycle $\beta \in \mathcal{T}_M$ with $\beta \neq c\,\mathrm{hol}_M^{(y)}$ for all $c \in \mathbb{R}$.

Proposition 5.1 states that the trajectories

$$\{\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M : t \geq 0\}$$

always equidistribute with respect to some invariant probability measure, even if the support of this measure does not contain the trajectory. Also, it implies the following.

**Corollary 5.2.** *Let $T \in \mathcal{H}(0)$ be a torus with no horizontal closed geodesics. For every $M \in \mathcal{F}_d(T)$ and every $\beta \in \mathcal{T}_M^{(0)}$, the surface $\mathrm{trem}_\beta\, M$ is generic for $\mu_T$ with respect to the real Rel flow.*

We will prove Proposition 5.1 and Corollary 5.2 in Section 5.4 pages 27 and 29 respectively.

5.2. **Proof of Theorem 1.2.** Without loss of generality, assume that $M$ has area 1. We can write

$$\mathrm{trem}_\beta\, M = \mathrm{trem}_{\beta'}\ u_{s_0} M,$$

where $s_0 = L_M(\beta)$ and

$$\beta' = \beta - s_0\ \mathrm{hol}_{u_{s_0}M}^{(y)} \in \mathcal{T}_{u_{s_0}M}^{(0)}.$$

Here we use the identification between $\mathcal{T}_M$ and $\mathcal{T}_{u_{s_0}M}$ ($\mathcal{T}_M^{(0)}$ and $\mathcal{T}_{u_{s_0}T}^{(0)}$) induced by the affine map corresponding to the action of $u_{s_0}$. By Corollary 5.2, we obtain

$$\eta_t^{\mathrm{trem}_\beta\, M} = \eta_t^{\mathrm{trem}_{\beta'}\, u_{s_0} M} \longrightarrow \mu_{u_{s_0}T}.$$

If $\beta \in \mathcal{T}_M$ is nontrivial and not a multiple of the cocycle $\mathrm{hol}_M^{(y)}$, then

$$\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M \notin \mathcal{E}(u_{s_0}T) \quad \text{for all } t \geq 0.$$

This follows from the fact that $\mathcal{E}(u_{s_0}T)$ is Rel-invariant, while the holonomy of the closed geodesics on $\mathrm{trem}_\beta\, M$ does not coincide with that of $M$.

5.3. **Properties of Rel on modular fibers.** We need the following results in preparation for the proof of Proposition 5.1 and Corollary 5.2.

By [Sch05, Theorem 2], the Veech group of $\mathcal{F}_d(T)$ is a lattice isomorphic to $A\,\mathrm{SL}(2,\mathbb{Z})A^{-1}$, where $A \in \mathrm{SL}(2,\mathbb{R})$ and $\Lambda = A\mathbb{Z}^2$ is the lattice defining $T$. Hence we have:

**Theorem 5.3.** *Let $T \in \mathcal{H}(0)$ be a torus, $d \geq 2$ be an integer and $\mu_T$ be the measure on $\mathcal{F}_d(T) \subset \mathcal{H}(1,1)$ which is induce by the finite covering map*

$$\Pi_T : \mathcal{F}_d(T) \longrightarrow T \setminus \{0\}.$$

*Then, the measure $\mu_T$ is finite, after rescaling assume it is a probability measure. Moreover, every trajectory of the real Rel flow in $\mathcal{F}_d(T)$ is either*

*(a) periodic. This occurs if and only if $T$ is horizontally periodic;*
*(b) dense and equidistributed with respect to the unique invariant probability measure $\mu_T$.*

*Proof.* By trajectories, we mean orbits that are defined for all times in an unbounded interval.

For a given $M \in \mathcal{F}_d(T)$, there exists an open interval

$$(-\varepsilon, \varepsilon) \subset \mathbb{R}$$

such that for all $t \in (-\varepsilon, \varepsilon)$, $\mathrm{Rel}_t M$ is defined. Since $\mathrm{Rel}_t M$ is obtained by a surgery that fixes the first singularity and moves the second singularity horizontally, by taking $\varepsilon > 0$ smaller if necessary we can find $d$-fold covering maps

$$\pi_t : \mathrm{Rel}_t M \to (T, 0, z+t)$$

such that

$$\{z + (t, 0) : t \in (-\varepsilon, \varepsilon)\} \subset T \setminus \{0\}$$

is a horizontal geodesic. This shows that

$$\mathrm{Rel}_t M \in \mathcal{F}_d(T) \quad \text{for all} \quad t \in (-\varepsilon, \varepsilon).$$

Since $M \in \mathcal{F}_d(T)$ is arbitrary and $\mathcal{F}_d(T)$ is closed, it follows that

$$\mathrm{Rel}_t M \in \mathcal{F}_d(T)$$

whenever $\mathrm{Rel}_t M$ is defined. Moreover, the real Rel trajectories coincide with the trajectories of the horizontal flow in $\mathcal{F}_d(T)$ with respect to the 1-form $\Pi_T^*(\mathrm{d}z)$. Since the Veech group of $\mathcal{F}_d(T)$ is a lattice by [Sch05, Theorem 2], the real Rel trajectories (horizontal trajectories) of all surfaces (points) are either periodic, or are dense and equidistribute with respect to a unique invariant probability measure $\mu_T$.

Let $\nu_T$ be the normalized Haar measure on $T$. Define

$$\mu'_T(U) := \frac{1}{\widetilde{d}} \int_{T\setminus\{0\}} \#(U \cap \Pi_T^{-1}(z))\, \mathrm{d}\nu_T(z),$$

where $\widetilde{d}$ is the degree of the covering map $\Pi_T : \mathcal{F}_d(T) \to T \setminus \{0\}$, see [EMS03, Corollary 3.3]. It follows that $\mu'_T$ defines a probability measure on $\mathcal{F}_d(T)$ and is invariant under the real Rel flow:

$$\begin{aligned}
\mu'_T(\mathrm{Rel}_t U) &= \frac{1}{\widetilde{d}} \int_{T\setminus\{0\}} \#(\mathrm{Rel}_t U \cap \Pi_T^{-1}(z))\, \mathrm{d}\nu_T(z) \\
&= \frac{1}{\widetilde{d}} \int_{T\setminus\{0\}} \#(\mathrm{Rel}_t U \cap \Pi_T^{-1}(\Upsilon_t z))\, \mathrm{d}\nu_T(z) \\
&= \frac{1}{\widetilde{d}} \int_{T\setminus\{0\}} \#(\mathrm{Rel}_t U \cap \mathrm{Rel}_t \Pi_T^{-1}(z))\, \mathrm{d}\nu_T(z) \\
&= \frac{1}{\widetilde{d}} \int_{T\setminus\{0\}} \#(U \cap \Pi_T^{-1}(z))\, \mathrm{d}\nu_T(z) \\
&= \mu'_T(U)
\end{aligned}$$

for all $t$ such that $\mathrm{Rel}_t U$ is defined. This implies that $\mu_T = \mu'_T$.

□

We call $\mu_T$ the *fiber probability measure*, that is, the normalized natural measure on $\mathcal{F}_d(T)$ given by Theorem 5.3. This measure is induced by the affine structure of the full Rel leaves.

The following result shows that according to $\mu_T$, the horizontal straight line flow on a generic surface $M \in \mathcal{F}_d(T)$ is uniquely ergodic, provided the torus $T$ has no horizontal closed geodesics.

**Theorem 5.4.** *Let $T \in \mathcal{H}(0)$ be an arbitrary torus and $d \geq 2$ be an integer. Let $\mu_T$ be the probability measure on $\mathcal{F}_d(T)$ of Theorem 5.3. Then,*

- *(a) If the horizontal straight-line flow on $T$ is minimal, then the horizontal flow on $\mu_T$-almost every surface $M \in \mathcal{F}_d(T)$ is uniquely ergodic.*
- *(b) The horizontal flow on a surface $M \in \mathcal{F}_d(T)$ is periodic if and only $T$ is horizontally periodic.*

*Proof.* The proof of (b) follows from covering arguments and the fact that the covering maps we are considering are translations in charts. Consequently, straight-line trajectories are preserved by the projection: a periodic trajectory in a given direction on the covering surface projects to a periodic trajectory in the same direction on the base.

We prove (a) as follows: we recall Masur's criterion for unique ergodicity, state Claim †, deduce (a) and finally prove Claim †.

Masur's criterion for unique ergodicity of the horizontal straight-line flow (see [AM24, Theorem 5.1.1]) states the following.

Let $M \in \mathcal{H}$ and let

$$\mathrm{proj} : \mathcal{H} \to \mathcal{M}_2$$

be the projection to the moduli space of Riemann surfaces of genus 2. Suppose that the Teichmüller geodesic

$$\{X_t = \mathrm{proj}(g_t M) : t > 0\}$$

is non-divergent, that is, there exist a sequence $t_n \to \infty$ and a compact set $C \subset \mathcal{M}_2$ such that

$$\mathrm{proj}(g_{-t_n} M) \in C.$$

If the horizontal flow on $M$ is minimal, then it is uniquely ergodic.

**Claim †.** *For every $\varepsilon > 0$ there exists a compact set*

$$K \subset \mathcal{H}(1,1)$$

*such that*

$$\mu_T\big(\{M \in \mathcal{F}_d(T) : \forall t_0 > 0\ \exists t \geq t_0,\ g_{-t}M \in K\}\big) > 1 - \varepsilon. \tag{15}$$

Now we prove Theorem 5.4, part (a). Let $\varepsilon > 0$ and let $K \subset \mathcal{H}(1,1)$ be as in Claim †. Then

$$\begin{aligned}\{M \in \mathcal{F}_d(T) : \forall t_0 > 0\ \exists t \geq t_0,\ g_{-t}M \in K\}\\ \subset \{M \in \mathcal{F}_d(T) : \text{the horizontal flow on } M \text{ is uniquely ergodic}\}.\end{aligned}$$

Indeed, if for all $t_0 > 0$ there exists $t \geq t_0$ such that $g_{-t}M \in K$, then

$$\mathrm{proj}(g_{-t}M) \in \mathrm{proj}(K) \subset \mathcal{M}_2.$$

Since $\mathrm{proj}(K)$ is compact, the trajectory $\{\mathrm{proj}(g_{-t}M) : t > 0\}$ is non-divergent in $\mathcal{M}_2$. By Masur's criterion, the horizontal straight-line flow on $M$ is uniquely ergodic.

By Claim †,

$$\mu_T(\{M \in \mathcal{F}_d(T) : \text{the horizontal flow on } M \text{ is uniquely ergodic}\}) > 1 - \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, (a) follows.

*Proof of Claim †.* The horizontal straight-line flow on $T \in \mathcal{H}(0)$ is minimal, then there exists a compact set

$$K_1 \subset \mathcal{H}(0)$$

such that for all $t_0$ there exists $t \geq t_0$ for which

$$g_{-t}T \in K_1.$$

Consider

$$K_2 := p^{-1}(F^{-1}(K_1)) \subset \mathcal{H}(1,1),$$

where $F$ and $p$ are the maps in eqs. (12) and (14) respectively. The set $K_2$ is closed but not compact. Define

$$C_\delta := \{N \in K_2 : \text{distance between the singularities is bigger than } \delta\}.$$

Then $C_\delta \subset \mathcal{H}(1,1)$ is compact for any $\delta > 0$. Since $K_1$ is compact, there exists a constant $\delta_1 > 0$ such that an Euclidean ball of radius $\delta_1$ can be immersed in every torus $T' \in K_1$. Let $\delta > 0$ satisfy

$$\delta < \min\{\sqrt{\varepsilon}\,\delta_1, \delta_1\}.$$

The compact set

$$K := C_\delta$$

satisfies eq. (15). Indeed,

$$\mu_T(\{M \in \mathcal{F}_d(T) : g_{-t}M \in K\})$$

is the proportion of points in $T$ whose images under the affine map $g_{-t}$ lie outside the Euclidean ball of radius $\delta$ centered at the origin in $g_{-t}T$. Hence

$$\frac{\mathrm{area}(T) - \pi\delta^2}{\mathrm{area}(T)} > 1 - \varepsilon.$$

Since for all $t_0 \geq 0$ there exists $t \geq t_0$ such that $g_{-t}T \in K_1$, the claim follows. △

□

**Corollary 5.5.** *For every $\varepsilon > 0$, there exists a compact set $K \subset \mathcal{F}_d(T)$ with $\mu_T(K) > 1-\varepsilon$ and $\mu_T(\partial K) = 0$.*

*Proof.* Since the map

$$\Pi_T : \mathcal{F}_d(T) \to T \setminus \{0\}$$

is a covering map of finite degree, the preimage $\Pi_T^{-1}(K')$ is compact whenever $K' \subset T \setminus \{0\}$ is compact.

Let $\nu_T$ be the normalized Haar measure on $T$. By definition,

$$\mu_T(\Pi_T^{-1}(H)) = 0 \qquad \text{whenever} \quad \nu_T(H) = 0, \quad H \subset T \setminus \{0\}.$$

Let $s$ be the injectivity radius of $T$, and let $\delta > 0$ be such that

$$0 < \delta \le \min\{\sqrt{\varepsilon}\, s, s\}.$$

Define $K' := T \setminus B(0, \delta)$ and $K := \Pi_T^{-1}(K')$, where $B(0, \delta)$ is the Euclidean open ball in $T$ of radius $\delta$ centered at the origin. Then $K \subset \mathcal{F}_d(T)$ is compact since $K'$ is compact.

Moreover, $\nu_T(\partial B(0, \delta)) = 0$ implies that $\mu_T(\partial K) = 0$. Since

$$\nu_T(B(0, \delta)) = \frac{\pi \delta^2}{\mathrm{area}(T)},$$

we have

$$\mu_T(K) = 1 - \nu_T(B(0, \delta)) > 1 - \varepsilon.$$

This proves the claim. □

5.4. **Proofs of Proposition 5.1 and Corollary 5.2.**

*Proof of Proposition 5.1.* We follow the same ideas in the proof of [CSW25, Theorem 1.5(iii)].

The steps for the proof are as follows: we first state Claim ‡, we indicate how this implies Proposition 5.1 and finally we prove Claim ‡.

**Claim ‡.** *For every $\varepsilon > 0$ and every $\delta > 0$, there exists $\ell_0 > 0$ such that for all $\ell \ge \ell_0$ there exists $A \subset [0, \ell]$ with $|A| > (1 - \varepsilon)\ell$ such that for all $t \in A$,*

$$\mathrm{dist}(\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M, \mathrm{Rel}_t\, u_{s_0} M) < \delta.$$

We now finish the proof of the Proposition 5.1.

Let $f \in C_c(\mathcal{H})$ be a continuous function with compact support. We will show that for all $\varepsilon' > 0$, there exists $\ell_3 > 0$ such that for all $\ell \ge \ell_3$ we have

$$\left| \int f \, \mathrm{d}\mu_{u_{s_0} T} - \frac{1}{\ell} \int_0^\ell f(\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M)\, \mathrm{d}t \right| < \varepsilon'. \tag{16}$$

Towards this, note that $\mathrm{supp}(f)$ is compact, so let $\delta > 0$ be such that if

$$\mathrm{dist}(M_1, M_2) < \delta,$$

then

$$|f(M_1) - f(M_2)| < \varepsilon'/3.$$

Applying Claim ‡ with

$$0 < \varepsilon < \frac{\varepsilon'}{6\, \|f\|_\infty}$$

and $\delta$, let $\ell_0 > 0$, such that for all $\ell \ge \ell_0$, there is a set $A \subset [0, \ell]$ satisfying

$$|A| > (1 - \varepsilon)\ell.$$

For all $\ell \ge \ell_0$,

$$I_\ell := \left| \frac{1}{\ell} \int_0^\ell f(\mathrm{Rel}_t\, u_{s_0} M)\, \mathrm{d}t - \frac{1}{\ell} \int_0^\ell f(\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M)\, \mathrm{d}t \right|$$

is bounded by

$$\frac{1}{\ell} \int_A |f(\mathrm{Rel}_t\, u_{s_0} M) - f(\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M)|\, \mathrm{d}t + \frac{2\, \|f\|_\infty\, \varepsilon \ell}{\ell}.$$

Hence, for all $\ell \ge \ell_0$

$$I_\ell < \frac{|A|}{\ell} \frac{\varepsilon'}{3} + 2\, \|f\|_\infty\, \varepsilon < \frac{2\varepsilon'}{3}.$$

By part (b) in Theorem 5.3, $u_{s_0}M$ is generic for $\mu_{u_{s_0}T}$ since $u_{s_0}T$ has no horizontal closed geodesics. Hence, there exists $\ell_3 \geq \ell_0$ such that for all $\ell \geq \ell_3$,

$$\left| \int f \, \mathrm{d}\mu_{u_{s_0}T} - \frac{1}{\ell} \int_0^\ell f(\mathrm{Rel}_t \, u_{s_0} M) \, \mathrm{d}t \right| < \frac{\varepsilon'}{3}.$$

By the triangle inequality, for all $\ell \geq \ell_3$, we obtain the inequality (16).

Finally, we have:

*Proof of claim ‡.* Let $Q \subset \mathcal{H} = \mathcal{H}(1,1)$ be a compact set such that there is $\ell_1 > 0$ such that for $\ell \geq \ell_1$, the set

$$A_1 = A_1(\ell) := \{t \in [0, \ell] : \mathrm{Rel}_t \, u_{s_0} M \in Q\}$$

satisfies

$$|A_1| > (1 - \varepsilon/2)\ell.$$

Such a $Q$ exists by Corollary 5.5.

Let $\widetilde{Q}$ be the compact of marked translation surfaces that map to $Q$.

By [CSW25, Proposition 4.8], the map

$$\{(\widetilde{q}, \beta) : \widetilde{q} \in \mathcal{H}_m, \ \beta \in \mathcal{T}_q^+\} \longrightarrow \mathcal{H}_m, \qquad (\widetilde{q}, \beta) \longmapsto \mathrm{trem}_\beta \, \widetilde{q}$$

is continuous.

Fix some norm $\|\cdot\|$ on $H^1(S, \Sigma; \mathbb{R}^2)$. For every $\delta > 0$, there exists $\delta' > 0$ such that for any

$$\widetilde{q} \in \widetilde{Q}$$

and $\beta_1, \beta_2 \in \mathcal{T}_q$ with

$$L_{\widetilde{q}}(\beta_1) = L_{\widetilde{q}}(\beta_2) = s_0,$$

if

$$\|\beta_1 - \beta_2\| < \delta',$$

then

$$\mathrm{dist}(\mathrm{trem}_{\beta_1} \widetilde{q}, \mathrm{trem}_{\beta_2} \widetilde{q}) < \delta.$$

Let $E \subset \mathcal{F}_d(u_{s_0}T)$ be the collection of surfaces with uniquely ergodic horizontal flow. By Theorem 5.4,

$$\mu_{u_{s_0}T}(E) = 1.$$

From [CSW25, Corollary 4.2], the set

$$\mathcal{C}_{+,s_0} := \{(M, \beta) : M \in \mathcal{H}, \ \beta \in C_M^+, \ L_M(\beta) = s_0\}$$

is closed. That is, if $(M_n, \beta_n) \in \mathcal{C}_{+,s_0}$ is a sequence such that

$$M_n \longrightarrow M \in \mathcal{H}, \qquad \beta_n \longrightarrow \beta \in H^1(S, \Sigma),$$

then $(M, \beta) \in \mathcal{C}_{+,s_0}$.

Also by [CSW25, Corollary 4.3], if $M \in \mathcal{H}$ has area 1 and the horizontal flow is uniquely ergodic. For every convergent sequence $(M_n, \beta_n) \in \mathcal{C}_{+,s_0}$ such that $M_n \to M$ we have

$$\beta_n \longrightarrow s_0 \, \mathrm{hol}_M^{(y)}.$$

The latter implies that for all $N \in E$ there exists $\alpha = \alpha(N) > 0$ such that if

$$N_1 \in \mathcal{H}, \qquad \mathrm{dist}(N_1, N) < \alpha$$

and

$$\beta_1 \in C_{N_1}^+, \qquad L_{N_1}(\beta_1) = s_0,$$

then

$$\left\| \beta_1 - s_0 \, \mathrm{hol}_N^{(y)} \right\| < \frac{\delta'}{2}.$$

By making $\alpha$ smaller if necessary, we can assume that

$$\left\| s_0 \, \mathrm{hol}_{N_1}^{(y)} - s_0 \, \mathrm{hol}_N^{(y)} \right\| < \delta'/2.$$

Denote by $V$ the open set

$$V := \{N_2 \in \mathcal{H}(1,1) : \exists N \in E, \ \mathrm{dist}(N_2, N) < \alpha(N)\}.$$

Clearly $E \subset V$, hence $V$ is a set of $\mu_{u_{s_0}T}$ full measure.

Since $u_{s_0}M$ is a generic point for $\mu_{u_{s_0}T}$, there exists $\ell_2 > 0$ such that for all $\ell \geq \ell_2$, the set

$$A_2 = A_2(\ell) := \{t \in [0, \ell] : \mathrm{Rel}_t\, u_{s_0}M \in V\}$$

satisfies

$$|A_2| > (1 - \varepsilon/2)\ell.$$

Let

$$\ell_0 = \max\{\ell_1, \ell_2\}.$$

For $\ell \geq \ell_0$, let

$$A := A_1(\ell) \cap A_2(\ell).$$

Then

$$|A| > (1 - \varepsilon)\ell.$$

Let $t \in A$ be arbitrary. For $\beta_t \in C^+_{\mathrm{Rel}_t M}$ such that

$$\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M = \mathrm{trem}_{\beta_t}\, \mathrm{Rel}_t\, M$$

we have

$$\left\| \beta_t - s_0\, \mathrm{hol}^{(y)}_{\mathrm{Rel}_t M} \right\| < \delta',$$

hence for a marking $\widetilde{q}'_t$ of $\mathrm{Rel}_t\, \mathrm{trem}_\beta\, M$ and a marking $\widetilde{q}_t$ of $\mathrm{Rel}_t\, u_{s_0}M$,

$$\mathrm{dist}(\widetilde{q}'_t, \widetilde{q}_t) < \delta.$$

△

□

*Proof of Corollary 5.2.* Through the proof we are assuming that $M$ has area 1. By Hahn decomposition, we can write

$$\beta = \beta^+ - \beta^-$$

for $\beta^\pm \in C^+_M$. We can assume

$$\beta^+ = \sum_{i=1}^{r^+} c_i \rho_i, \qquad \beta^- = \sum_{i=r^++1}^{r^++r^-} c_i \rho_i,$$

where $r^+$ and $r^-$ are non-negative integers with for $\rho_i \in C^{+,erg}_M$ such that $L_M(\rho_i) = 1$ and $c_i \geq 0$ for all $i = 1, \ldots, r^+ + r^-$. For simplicity, let $r$ be the sum $r^+ + r^-$. The foliation cocycles $\rho_1, \ldots, \rho_r$ correspond to all the invariant ergodic probability measures for the horizontal flow on $M$, hence we can write

$$\mathrm{hol}^{(y)}_M = \sum_{i=1}^{r} \frac{1}{r} \rho_i.$$

Define $s_1 = r \max\{c_{r^++1}, \ldots, c_r\}$, thus

$$s_1\, \mathrm{hol}^{(y)}_M + \beta = \sum_{i=1}^{r^+} \left(\frac{s_1}{r} + c_i\right) \rho_i + \sum_{i=r^++1}^{r} \left(\frac{s_1}{r} - c_i\right) \rho_i.$$

For all $i = r^+ + 1, \ldots, r$ we have $\frac{s_1}{r} - c_i \geq 0$ then

$$\beta' = s_1\, \mathrm{hol}^{(y)}_M + \beta \in C^+_M.$$

The affine map corresponding to $u_{-s_1}$ identifies the foliation cocycles on $C^+_M$ with $C^+_{u_{-s_1}M}$, so we can think of $\beta' \in C_{u^+_{-s_1}}$ and write

$$\mathrm{trem}_\beta\, M = \mathrm{trem}_{\beta'}\, u_{-s_1}M. \tag{17}$$

Let $M' = u_{-s_1}M \in \mathcal{F}_d(T')$ with $T' = u_{-s_1}T$ and $\beta' \in C^+_{M'}$ be as before. By linearity, $L_{M'}(\beta') = s_1$, thus by Proposition 5.1 the Rel trajectory of $\mathrm{trem}_{\beta'}\, M'$ is generic for $\mu_{u_{s_1}T'} = \mu_T$. By eq. (17), the Rel trajectory of $\mathrm{trem}_\beta\, M$ is generic for $\mu_T$.

□

## 6. Orbit closures

Throughout this work, we assume that $T \in \mathcal{H}(0)$ is a flat torus with no horizontal closed geodesics. This assumption will be stated explicitly when needed, but may be omitted in intermediate calculations. In particular, every surface in $\mathcal{E}(T) := \mathcal{F}_2(T)$ has aperiodic horizontal flow.

We classify surfaces in $\mathcal{E}(T)$ according to the horizontal flow into three types: non-minimal ($tor$), minimal non-uniquely ergodic ($m$-$nue$) and minimal uniquely ergodic ($min$), corresponding to $\mathcal{E}^{(tor)}(T)$, $\mathcal{E}^{(m-nue)}(T)$, and $\mathcal{E}^{(min)}(T)$.

Explicitly,

$$\mathcal{E}^{(tor)}(T) = \{T\#_I T : \mathrm{hol}_T(I) = (x, 0),\ |x| > 0\},$$

$$\mathcal{E}^{(m-nue)}(T) = \{M \in \mathcal{E}(T) : M \text{ has minimal non-uniquely ergodic horizontal flow}\}$$

and

$$\mathcal{E}^{(min)}(T) = \mathcal{E}(T) \setminus \left(\mathcal{E}^{(tor)}(T) \cup \mathcal{E}^{(m-nue)}(T)\right).$$

Let $a \geq 0$ and define

$$\mathcal{SU}_{\leq a}(T) := \left\{\mathrm{trem}_{\beta'} M' : M' \in \mathcal{E}(T),\ \beta' \in \mathcal{T}_{M'}^{(0)},\ |L|_{M'}(\beta') \leq a\right\}.$$

Similar sets were introduced in [CSW25] under the name of "spiky fish" to describe orbit closures of the horocycle flow. We use the notation $\mathcal{SU}$ for *sea urchin*, as these sets are more restricted than the spiky fish and the tremored surfaces in $\mathcal{SU}_{\leq a}(T) \setminus \mathcal{E}(T)$ resemble the spikes of a sea urchin.

We devote this section to proving the following result (see page 31).

**Proposition 6.1.** *Assume that $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. Let $M \in \mathcal{E}(T)$ be a translation surface whose horizontal flow is not ergodic with respect to* Leb *and has no horizontal closed geodesics. Let $\beta \in \mathcal{T}_M^{(0)}$ be a balanced foliation cocycle with total variation $|L|_M(\beta) = a \geq 0$. If $\mathrm{Rel}_t M$ is defined for all $t \geq 0$, then*

$$\overline{\{\mathrm{Rel}_t\, \mathrm{trem}_\beta M : t \geq 0\}} = \mathcal{SU}_{\leq a}(T).$$

Dropping the requirement of balanced tremors, Proposition 6.1 shows that, whenever $\mathcal{E}(T)$ contains surfaces whose horizontal flow is minimal but not uniquely ergodic, real Rel trajectories of a tremored surface are dense among tremored surfaces in the fiber $\mathcal{E}(T)$ with no greater variation.

Note that when $\beta = 0$, we recover the density statement in item (2) of Theorem 3.1 since $\mathcal{SU}_{\leq 0}(T) = \mathcal{E}(T)$.

As a consequence of Proposition 6.1, we obtain the following.

**Corollary 6.2.** *Assume that $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. Let $M \in \mathcal{E}(T)$ and $\beta' \in \mathcal{T}_M$. Then*

$$\overline{\{\mathrm{Rel}_t\, \mathrm{trem}_{\beta'} M : t \geq 0\}} = \mathcal{SU}_{\leq a'}(u_{s_0} T),$$

*where $s_0 = L_M(\beta')$ and $a' = |L|_M(\beta' - s_0\, \mathrm{hol}_M^{(y)})$.*

### 6.1. Proofs of Theorem 1.1 and Corollary 6.2.

*Proof of Theorem 1.1.* Let $M \in \mathcal{E}^{(m-nue)}(T)$ and $\beta \in \mathcal{T}_M$. By Corollary 6.2, we obtain $T' = u_{s_0} T$ and

$$a' = |L|_M(\beta - s_0\, \mathrm{hol}_M^{(y)}) \geq 0$$

such that

$$\overline{\{\mathrm{Rel}_t\, \mathrm{trem}_\beta M : t \geq 0\}} = \mathcal{SU}_{\leq a'}(T').$$

Since both $\mathrm{Rel}_t M$ and $\mathrm{Rel}_t\, \mathrm{trem}_\beta M$ are defined for all $t \in \mathbb{R}$, we have

$$\{\mathrm{Rel}_t\, \mathrm{trem}_\beta M : t \in \mathbb{R}\} \subset \mathcal{SU}_{\leq a'}(T').$$

The equality after taking the closure on the left follows from the fact that $\mathcal{SU}_{\leq a'}(T')$ is closed and

$$\{\mathrm{Rel}_t\, \mathrm{trem}_\beta M : t \geq 0\} \subset \{\mathrm{Rel}_t\, \mathrm{trem}_\beta M : t \in \mathbb{R}\}.$$

Moreover, if $a' \neq 0$, the holonomy of the closed geodesics on $\mathrm{trem}_\beta M$ differs from that on $M$. This implies that

$$\mathcal{SU}_{=0}(T') = \mathcal{E}(T') \subsetneq \mathcal{SU}_{\leq a'}(T').$$

$\mathcal{SU}_{\leq a'}(T')$ is not $\mathrm{SL}(2, \mathbb{R})$-invariant. For instance,

$$g_h\, \mathrm{trem}_\beta M = \mathrm{trem}_{e^h \beta}(g_h M),$$

and in general $g_h M \notin \mathcal{E}(T')$. Moreover,

$$|L|_{g_h M}(e^h \beta) = e^h |L|_M(\beta).$$

Thus, for all sufficiently large $h > 0$, we have $|L|_{g_h M}(e^h \beta) > a'$. Hence the $\mathrm{SL}(2,\mathbb{R})$-orbit of $\mathrm{trem}_\beta M$ is not contained in $\mathcal{SU}_{\leq a'}(T')$.

□

*Proof of Corollary 6.2.* By Lemma 4.2, $\beta = \beta' - c\,\mathrm{hol}_M^{(y)} \in \mathcal{T}_M^{(0)}$. Proposition 6.1 implies that

$$\overline{\{\mathrm{Rel}_t \,\mathrm{trem}_\beta M : t \geq 0\}} = \mathcal{SU}_{\leq a'}(T).$$

For every $s \in \mathbb{R}$, the map $u_s$ induces an identification $\mathcal{T}_M \simeq \mathcal{T}_{u_s M}$, see [CSW25, § 6]. The map $u_s$ is a homeomorphism that commutes with Rel and trem, that is $\mathrm{Rel}_t u_s M = u_s \mathrm{Rel}_t M$ and $\mathrm{trem}_\beta u_s M = u_s \mathrm{trem}_\beta M$.

Moreover, the foliation cocycle $s\,\mathrm{hol}_M^{(y)}$ satisfies

$$\mathrm{trem}_{s\,\mathrm{hol}_M^{(y)}} M = u_s M$$

and $L_{u_s M}(\beta) = L_M(\beta)$. This implies

$$u_{s_0}\left(\overline{\{\mathrm{Rel}_t \,\mathrm{trem}_\beta M : t \geq 0\}}\right) = \overline{\{\mathrm{Rel}_t \,\mathrm{trem}_{\beta'} M : t \geq 0\}} \tag{18}$$

and

$$u_{s_0} \mathcal{SU}_{\leq b}(T) = \mathcal{SU}_{\leq b}(u_{s_0} T), \qquad b \geq 0. \tag{19}$$

By combining eq. (18) and eq. (19), we obtain the result. □

6.2. **Proof of Proposition 6.1.** The proof of Proposition 6.1 requires two intermediate results: Proposition 6.3 and Proposition 6.4, which we first state.

**Proposition 6.3.** *Assume $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. Let $M' \in \mathcal{E}^{(m-nue)}(T) \cup \mathcal{E}^{(tor)}(T)$ and $\beta' \in \mathcal{T}_{M'}^{(0)}$ with $|L|_{M'}(\beta') = a$. Then for every $M \in \mathcal{E}^{(m-nue)}(T) \cup \mathcal{E}^{(tor)}(T)$, there exist a sequence $s_j \to \infty$ such that for every $\beta \in \mathcal{T}_M^{(0)}$ with $|L|_M(\beta) = a$ we have*

$$\mathrm{Rel}_{s_j} \mathrm{trem}_\beta M \longrightarrow \mathrm{trem}_{\beta'} M'.$$

The proof of Proposition 6.3 (page 33) is the focus of Section 6.4.

**Proposition 6.4.** *Let $T \in \mathcal{H}(0)$ be a torus such that $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. Let $M \in \mathcal{E}^{(tor)}(T) \cup \mathcal{E}^{(m-nue)}(T)$ and $\beta \in \mathcal{T}_M^{(0)}$. If $M \in \mathcal{E}^{(tor)}(T)$, assume that its real Rel trajectory is defined for all positive times. Then*

$$\overline{\{\mathrm{Rel}_t \,\mathrm{trem}_\beta M : t \geq 0\}} \supset \{\mathrm{trem}_{h\beta} M : h \in [0,1]\}.$$

We will deduce Proposition 6.4 (page 38) using Lemma 6.8. Section 6.5 treats these statements.

**Remark 6.5.** The assumption $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$ is necessary. Assume that $T$ is normalized as in Lemma 3.5 with $\alpha \in (0,1)$ irrational. By Theorem 3.4, $\alpha$ has bounded partial quotients if and only if $\mathcal{E}^{(m-nue)}(T) = \emptyset$.

By [Osp24, Theorem 1.1], $\alpha$ has bounded partial quotients if and only if the real Rel trajectories are not recurrent. More precisely, for every $M \in \mathcal{E}^{(tor)}(T)$ and $\beta \in C_M^+ \setminus \{0\}$, there exist an open set $U$ containing $\mathrm{trem}_\beta M$ and $\widetilde{t} > 0$ such that

$$\mathrm{Rel}_t \,\mathrm{trem}_\beta M \notin U \quad \text{for all } t \geq \widetilde{t}.$$

This implies that there exists some $1 > h' > 0$ such that

$$\overline{\left\{\mathrm{Rel}_t \,\mathrm{trem}_\beta M : t \geq \widetilde{t}\,\right\}} \cap \{\mathrm{trem}_{h\beta} M : h \in (h',1)\} = \emptyset.$$

*Proof of Proposition 6.1.* Let $M \in \mathcal{E}(T)$ and $\beta \in \mathcal{T}_M^{(0)}$ be fixed as in the the hypothesis. Denote the trajectory of $\mathrm{trem}_\beta M$ by

$$\mathcal{O}(\mathrm{trem}_\beta M) := \{\mathrm{Rel}_t \,\mathrm{trem}_\beta M : t \geq 0\}.$$

By Proposition 6.3, we obtain

$$\overline{\mathcal{O}(\mathrm{trem}_\beta M)} \supset \mathcal{SU}_{=a}(T),$$

where

$$\mathcal{SU}_{=a}(T) := \left\{\mathrm{trem}_{\beta'} M' : M' \in \mathcal{E}(T),\ \beta' \in \mathcal{T}_{M'}^{(0)},\ |L|_{M'}(\beta') = a\right\}.$$

By Proposition 6.4, we also have

$$\overline{\mathcal{O}(\operatorname{trem}_\beta M)} \supset \{\operatorname{trem}_{h\beta} M : h \in [0,1]\}. \tag{20}$$

The real Rel flow is continuous whenever it is defined: if $N_i \to N$ and $\operatorname{Rel}_s N$ is defined, then for all sufficiently large $i$, $\operatorname{Rel}_s N_i$ is defined and

$$\operatorname{Rel}_s N_i \to \operatorname{Rel}_s N \quad \text{as } i \to \infty.$$

It implies that if $N \in \overline{\mathcal{O}(\operatorname{trem}_\beta M)}$, then $\operatorname{Rel}_s N \in \overline{\mathcal{O}(\operatorname{trem}_\beta M)}$ whenever defined. Particularly, for all $h \geq 0$ and for all $t \geq 0$, $\operatorname{Rel}_t \operatorname{trem}_{h\beta} M$ is defined.

Since $|L|_M(h\beta) = ha$, combining eq. (20) with Proposition 6.3 (replacing $\beta$ with $h\beta$ and $a$ by $a' = ha$), we obtain

$$\overline{\mathcal{O}(\operatorname{trem}_\beta M)} \supset \bigcup_{a' \in [0,a]} \mathcal{SU}_{=a'}(T) = \mathcal{SU}_{\leq a}(T). \tag{21}$$

On the other hand, it is immediate that

$$\mathcal{O}(\operatorname{trem}_\beta M) \subset \mathcal{SU}_{\leq a}(T). \tag{22}$$

Since $\mathcal{SU}_{\leq a}(T)$ is closed, combining Equation (21) and Equation (22) yields the result. ☐

6.3. **Area exchange to control tremors.** We study how the real Rel trajectory of a tremored surface changes its period coordinates for surfaces in $\mathcal{E}$. These changes can be computed explicitly.

We first recall the following result.

**Proposition 6.6** ([CSW25, Proposition 3.2 and Lemma 10.5])**.** *Let $T \in \mathcal{H}(0,0)$ be a torus with two marked points. Let $I_1$ and $I_2$ be two saddle connections in $T$. Then*

$$T\#_{I_1}T = T\#_{I_2}T \in \mathcal{E}(T)$$

*if and only if the corresponding homology classes $[I_1]$ and $[I_2]$ are equal as elements of*

$$H_1\left(T, \Sigma; \mathbb{Z}/2\mathbb{Z}\right).$$

*The saddle connections $I_1$ and $I_2$ partition $T$ into the components of $T \setminus (I_1 \cup I_2)$, which admit a two-coloring. No two adjacent components share the same color.*

Let $A_1$ and $A_2$ be the two-colored regions in Proposition 6.6. We define the *area exchange* associated to the segments $I_1$ and $I_2$ as

$$\frac{|\operatorname{area}(A_1) - \operatorname{area}(A_2)|}{\operatorname{area}(T)} = \frac{|\operatorname{area}(T) - 2\operatorname{area}(A_1)|}{\operatorname{area}(T)},$$

since $\operatorname{area}(A_1) + \operatorname{area}(A_2) = \operatorname{area}(T)$.

Let $M \in \mathcal{E}(T)$ and suppose that $T^1\#_I T^2$ is a marked representation. In period coordinates we write

$$\left(\operatorname{hol}_T(\zeta_1^1), \operatorname{hol}_T(\zeta_2^1), \operatorname{hol}_T(\zeta_1^2), \operatorname{hol}_T(\zeta_2^2), \operatorname{hol}_T(I)\right) \in (\mathbb{R}^2)^5,$$

where $\zeta_1^i, \zeta_2^i$ generate $H_1(T^i)$ seen as elements of $H_1(M, \Sigma)$ and $I \subset T$ is a geodesic segment. Then

$$\{\zeta_1^i : i = 1,2\} \cup \{\zeta_2^i : i = 1,2\} \cup \{I\}$$

generate $H_1(M, \Sigma)$.

Assume that $M_t = \operatorname{Rel}_t M$ is defined for some $t \in \mathbb{R}$. Then $M_t$ admits two representations

$$M_t = T\#_{I_t}T \tag{23}$$

and

$$M_t = T\#_{I'_t}T. \tag{24}$$

In a marking corresponding to (23), the period coordinates are

$$\left(\operatorname{hol}_T(\zeta_1^1), \operatorname{hol}_T(\zeta_2^1), \operatorname{hol}_T(\zeta_1^2), \operatorname{hol}_T(\zeta_2^2), \operatorname{hol}_T(I) + (t,0)\right), \tag{25}$$

since real Rel acts by adding $(t,0)$ to the slit coordinate.

In a marking corresponding to (24), the period coordinates are

$$\big(\mathrm{hol}_T(\zeta_1^1), \mathrm{hol}_T(\zeta_2^1), \mathrm{hol}_T(\zeta_1^2), \mathrm{hol}_T(\zeta_2^2), \mathrm{hol}_T(I'_t)\big), \tag{26}$$

with

$$\mathrm{hol}_T(I) + (t, 0) - \mathrm{hol}_T(I'_t) \in 2\Lambda,$$

where $\Lambda$ is the lattice defining $T$.

The study of the period coordinates of $\mathrm{Rel}_t\,\mathrm{trem}_\beta\, M$ is more subtle. Intuitively, as $t \to \infty$, the segment $I_t$ becomes longer and its direction tends to the horizontal. Since the horizontal flow on $T$ is minimal, the values of $\mathrm{hol}_{\mathrm{Rel}_t\,\mathrm{trem}_\beta\, M}(\zeta_k^i)$ become asymptotically proportional to $\mathrm{hol}_{\mathrm{trem}_\beta\, M}(\zeta_k^i)$, with proportionality by the area exchange associated with $I_t$ and $I'_t$.

6.4. **Proof of Proposition 6.3.** We prove an approximation result along real Rel trajectories with control on the associated foliation cocycles.

**Lemma 6.7.** *Let $T \in \mathcal{H}(0)$ be as above and assume $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. For every $M, M' \in \mathcal{E}^{(m-nue)}(T) \cup \mathcal{E}^{(tor)}(T)$ there exist $s_j \to \infty$ such that for every $\beta' \in C^+_{M'}$ with $L_{M'}(\beta') = 1$, there exists $\widetilde{\beta} \in C^+_M$ with $L_M(\widetilde{\beta}) = 1$ satisfying*

$$\mathrm{Rel}_{s_j} M \longrightarrow M', \qquad \left\| \widetilde{\beta}_j - \beta' \right\| \longrightarrow 0,$$

*where $\widetilde{\beta}_j$ is defined by*

$$\mathrm{trem}_{\widetilde{\beta}_j} \mathrm{Rel}_{s_j} M = \mathrm{Rel}_{s_j} \mathrm{trem}_{\widetilde{\beta}} M.$$

*Hence* $\mathrm{Rel}_{s_j}\,\mathrm{trem}_{\widetilde{\beta}}\, M \to \mathrm{trem}_{\beta'}\, M'$.

We begin by deducing Proposition 6.3. The proof of Lemma 6.7 is deferred to page 33.

*Proof of Proposition 6.3.* Recall that in the hypothesis, $\beta'$ is a balanced foliation cocycle. By Lemma 4.2, we can write

$$\beta' = \beta'^{+} - \beta'^{-}$$

with $\beta'^{+}, \beta'^{-} \in C^{+,erg}_{M'}$. Then $L_{M'}(\beta'^{+}) = L_{M'}(\beta'^{-}) = a/2$. Let $s_j$ be given by Lemma 6.7. Then there exist distinct $\beta^+, \beta^- \in C^{+,erg}_M$ such that $\beta = \beta^+ - \beta^- \in \mathcal{T}^{(0)}_M$ and $|L|_M(\beta) = a$, with

$$\mathrm{Rel}_{s_j} M \to M', \qquad \|\beta - \beta'\| \to 0.$$

Hence $\mathrm{Rel}_{s_j}\,\mathrm{trem}_\beta\, M \to \mathrm{trem}_{\beta'}\, M'$.

The involution $\iota : M \to M$ is a translation equivalence and induces a continuous map on $\mathcal{T}^{(0)}_M$. Additionally,

$$\{\beta'' \in \mathcal{T}^{(0)}_M : |L|_M(\beta'') = a\} = \{\beta, \iota^*\beta\}.$$

If $\beta \in \mathcal{T}^{(0)}_M$ and $s_j \to \infty$ are such that

$$\mathrm{Rel}_{s_j}\,\mathrm{trem}_\beta\, M \to \mathrm{trem}_{\beta'}\, M',$$

then

$$\mathrm{Rel}_{s_j}\,\mathrm{trem}_{\iota^*\beta}\, M \to \mathrm{trem}_{\iota^*\beta'}\, M'.$$

Finally, as translation surfaces, $\mathrm{trem}_{\beta'}\, M'$ and $\mathrm{trem}_{\iota^*\beta'}\, M'$ coincide. □

*Proof of Lemma 6.7.* Without loss of generality, we may assume that $T \in \mathcal{H}(0)$ has area one and is normalized as in Section 3.2 and Lemma 3.5, with $\alpha$ irrational. Then $M$ and $M'$ have area two. We may assume $\beta' \in C^{+,erg}_{M'}$ and we will find $\widetilde{\beta} \in C^{+,erg}_M$ as required.

Since $\alpha$ has unbounded partial quotients (by Theorem 3.4 and Remark 3.6), let $q_{i_j}$ be a subsequence of denominators of best approximants such that $a_{i_j+1} \to \infty$. By Lemma 4.3 we may take $t_j = \log q_{i_j}$.

Denote $T_j = g_{-t_j} T$. Let $\varepsilon > 0$ be small and define markings

$$T_j^1 \#_{F_j} T_j^2 \quad \text{of } g_{-t_j} M \text{ with } \mathrm{hol}_{T_j}(F_j) = \left( \frac{u_j}{q_{i_j}}, v_j q_{i_j} \right),$$

$$T_j^1 \#_{G_j} T_j^2 \quad \text{of } g_{-t_j} M' \text{ with } \mathrm{hol}_{T_j}(G_j) = \left( \frac{x_j}{q_{i_j}}, y_j q_{i_j} \right),$$

where

$$|x_j|/q_{i_j} + |y_j|q_{i_j} < \varepsilon \quad \text{and} \quad |u_j|/q_{i_j} + |v_j|q_{i_j} < \varepsilon$$

for all $j$ such that $t_j \geq \tilde{t}$ as in Lemma 4.3 and Theorem 4.5. Let $\tilde{j} > 0$ such that $t_j \geq \tilde{t}$ for all $j \geq \tilde{j}$. If $M' \in \mathcal{E}^{(tor)}(T)$, assume $x_j = x \neq 0$ and $y_j = 0$ for all $j$. Likewise, if $M \in \mathcal{E}^{(tor)}(T)$, assume $u_j = u > 0$ and $v_j = 0$ for all $j$.

For $j$ sufficiently large, define

$$s_j = 2q_{i_j} - u_j + x_j.$$

Then $s_j \to \infty$ as $j \to \infty$. Indeed, for every $\varepsilon \in (0, 1/2)$ and increasing $\tilde{j}$ if necessary,

$$0 < 2(1-\varepsilon)q_{i_j} < s_j < 2(1+\varepsilon)q_{i_j} \quad \text{for all} \quad j \geq \tilde{j}.$$

Observe that a marking for $\mathrm{Rel}_{s_j/q_{i_j}}\ g_{-t_j} M$ is of the form

$$T_j^1 \#_{J_j} T_j^2 = T_j^1 \#_{J_j'} T_j^2,$$

where

$$\mathrm{hol}_{T_j}(J_j) = \left(2 + \frac{x_j}{q_{i_j}}, v_j q_{i_j}\right), \qquad \mathrm{hol}_{T_j}(J_j') = \left(\frac{x_j}{q_{i_j}}, v_j q_{i_j} + (-1)^{i_j+1} 2q_{i_j} \left\|q_{i_j}\alpha\right\|\right).$$

By eq. (10) and Proposition 6.6,

$$\mathrm{hol}_{T_j}(J_j) - \mathrm{hol}_{T_j}(J_j') \in 2\Lambda_{i_j},$$

which determines the holonomy of $J_j'$.

The segments $J_j$ and $J_j'$ intersect once at their midpoints, forming a parallelogram. The area of this parallelogram is

$$b_j = \frac{\left(2 + \frac{x_j}{q_{i_j}}\right) q_{i_j} \left\|q_{i_j}\alpha\right\|}{2}.$$

Since $\frac{x_j}{q_{i_j}} \to 0$, $|x_j| \left\|q_{i_j}\alpha\right\| < \frac{|x_j|}{q_{i_j+1}}$ and $q_{i_j} \left\|q_{i_j}\alpha\right\| < \frac{1}{a_{i_j+1}}$, we obtain $b_j \to 0$ as $j \to \infty$.

A marking of $\mathrm{Rel}_{s_j} M$ is given by

$$T^1 \#_{H_j} T^2 = T^1 \#_{H_j'} T^2,$$

where

$$\mathrm{hol}_T(H_j) = (2q_{i_j} + x_j, v_j), \qquad \mathrm{hol}_T(H_j') = \left(x_j, v_j + (-1)^{i_j+1} 2 \left\|q_{i_j}\alpha\right\|\right),$$

via the affine map induced by $g_{t_j}$. Since this map preserves area, the parallelogram formed by $H_j$ and $H_j'$ is $b_j$.

The markings of $M'$ and $\mathrm{Rel}_{s_j} M$, denoted by $T^1 \#_{I_j} T^2$ and $T^1 \#_{H_j'} T^2$, have comparable period coordinates. They share the same coordinates from $T$, and the slit holonomies have identical $x$-coordinates:

$$\mathrm{hol}_T(I_j) = (x_j, y_j), \qquad \mathrm{hol}_T(H_j') = \left(x_j, v_j + (-1)^{i_j+1} 2 \left\|q_{i_j}\alpha\right\|\right). \tag{27}$$

The $y$-coordinates differ by a quantity tending to zero, hence $\mathrm{Rel}_{s_j} M \to M'$, as $j \to \infty$.

For $M'$, let $\mu_j' = \mathrm{Leb}|_{A_j'}$ be the probability measure for the flow in the direction of $I_j$ obtained by restricting Leb to a connected component $A_j' \subset M' \setminus (I_j \cup \iota(I_j))$.

Let $\nu_j'$ be the corresponding transverse measure and $\beta_j'$ the associated foliation cocycle. By Theorem 4.5, after increasing $\tilde{j}$ and replacing $A_j'$ by $\iota(A_j')$ if necessary, we have

$$\left\|\beta' - \beta_j'\right\| < \varepsilon/4 \quad \text{for all} \quad j \geq \tilde{j}.$$

If $M' \in \mathcal{E}^{(tor)}(T)$, then $\beta_j' = \beta'$ for all $j$, since $I_j = I$.

For $\mathrm{Rel}_{s_j} M$, let $\widehat{C}_j \subset \mathrm{Rel}_{s_j} M \setminus (H_j' \cup \iota(H_j'))$ be a connected component and set $\hat{\mu}_j = \mathrm{Leb}|_{\widehat{C}_j}$, an invariant probability measure for the flow on $\mathrm{Rel}_{s_j} M$ in the direction of $H_j'$. Let $\widehat{\nu}_j$ be the corresponding transverse measure. The period coordinates given by the markings on $M'$ and $\mathrm{Rel}_{s_j} M$ differ only by the vectors in eq. (27). Since $\mu_j'$ and $\widehat{\mu}_j$ are both restrictions of Leb in charts, by increasing $\tilde{j}$ and replacing $\widehat{C}_j$ by $\iota(\widehat{C}_j)$ if necessary, we obtain that

$$\left\|\beta_j' - \beta_{\widehat{\nu}_j}\right\| < \varepsilon/4 \quad \text{for all} \quad j \geq \tilde{j}.$$

The area contained by the parallelogram between $H_j$ and $H'_j$ can be made arbitrarily close to 0 by increasing $\widetilde{j}$. Thus, let $B_j \subset \mathrm{Rel}_{s_j} M \backslash (H_j \cup \iota(H_j))$ be a connected component such that $\mathrm{Leb}(B_j \cap \widehat{C}_j) = 1 - b_j$. Set $\mu_j = \mathrm{Leb}\,|_{B_j}$ and $\nu_j$ the corresponding transverse measure. By making $\widetilde{j}$ larger, we can assume that

$$\left\|\beta_{\nu_j} - \beta_{\widehat{\nu}_j}\right\| < \varepsilon/4 \quad \text{for all} \quad j \geq \widetilde{j}.$$

Since

$$\mathrm{hol}^{(y)}_{\mathrm{Rel}_{s_j} M}(H_j), \ \mathrm{hol}^{(y)}_{\mathrm{Rel}_{s_j} M}(H'_j) \longrightarrow 0,$$

the cocycles $\beta_{\nu_j}$ and $\beta_{\widehat{\nu}_j}$ are close to foliation cocycle $\widetilde{\beta}_j \in C^{+,erg}_{\mathrm{Rel}_{s_j} M}$ for the horizontal flow on $\mathrm{Rel}_{s_j} M$, that is

$$\left\|\beta_{\nu_j} - \widetilde{\beta}_j\right\| < \varepsilon/4 \quad \text{for all} \quad j \geq \widetilde{j}$$

with

$$L_{\mathrm{Rel}_{s_j} M}(\widetilde{\beta}_j) = 1.$$

Moreover, $\widetilde{\beta}_j$ corresponds to a foliation cocycle on $M$, which we denote by $\widetilde{\beta}_j$ as well, via the natural identification

$$C^{+,erg}_{M} \simeq C^{+,erg}_{\mathrm{Rel}_{s_j} M}.$$

That is,

$$\mathrm{trem}_{\widetilde{\beta}_j} \mathrm{Rel}_{s_j} M = \mathrm{Rel}_{s_j} \mathrm{trem}_{\widetilde{\beta}_j} M.$$

Since $L_M(\widetilde{\beta}_j) = L_{\mathrm{Rel}_{s_j} M}(\widetilde{\beta}_j) = 1$ and $C^{+,erg}_M$ is spanned by two foliation cocycles, after passing to a subsequence we may assume that $\widetilde{\beta}_j$ is constant. Denote this common value by $\widetilde{\beta}$.

Finally, by the triangle inequality,

$$\left\|\beta' - \widetilde{\beta}_j\right\| < \varepsilon \quad \text{for all } j \geq \widetilde{j}.$$

By continuity of the tremor map, for all sufficiently large $j$, a marking of

$$\mathrm{Rel}_{s_j} \mathrm{trem}_{\beta} M$$

is arbitrarily close to a marking of $\mathrm{trem}_{\beta'} M'$, and the proof is complete. □

6.5. **Proof of Proposition 6.4.** We defer the proof of Proposition 6.4 to page 38, since it depends on the following result.

**Lemma 6.8.** *Let $T \in \mathcal{H}(0)$ be a torus with $\mathcal{E}^{(m-nue)}(T) \neq \emptyset$. Let $M \in \mathcal{E}^{(tor)}(T) \cup \mathcal{E}^{(m-nue)}(T)$. If $M \in \mathcal{E}^{(tor)}(T)$, assume its real Rel trajectory is defined for all positive times.*

*For every $h \in [0,1]$, there exists a sequence $s_j \to \infty$ as $j \to \infty$ satisfying*

*(1)* $\mathrm{Rel}_{s_j} M \longrightarrow M$.
*(2) For each $j$ we can represent $\mathrm{Rel}_{s_j} M$ in two ways,*

$$T \#_{F_j} T = T \#_{F'_j} T,$$

*such that the corresponding area exchange $h_j$ satisfies $h_j \to h$.*
*(3) For each $j$ the direction of the segments $F_j$, $F'_j \subset T$ converges to the horizontal direction and as $j \to \infty$*

$$\mathrm{hol}^{(x)}_T(F_j), \ \mathrm{hol}^{(x)}_T(F'_j) \to \infty,$$
$$\mathrm{hol}^{(y)}_T(F_j), \ \mathrm{hol}^{(y)}_T(F'_j) \to 0.$$

*(4) We can write $M = T \#_{I_j} T$ with $\mathrm{hol}^{(x)}_T(I_j) = \mathrm{hol}^{(x)}_T(F'_j)$ and*

$$\lim_{j \to \infty} |\,\mathrm{hol}^{(y)}_T(I_j) - \mathrm{hol}^{(y)}_T(F'_j)| = 0.$$

*Where the sequence $\{I_j\}$ is as in Lemma 4.3 and Theorem 4.5.*

*Proof.* Without loss of generality, we assume that $T \in \mathcal{H}(0)$ has area one and is normalized as in Section 3.2 and Lemma 3.5, where $\alpha \in (0,1)$ is irrational.

Write $\alpha = [a_1, a_2, \dots]$ with unbounded partial quotients by Theorem 3.4 and Remark 3.6.

Choose a subsequence $\{i_j\}$ with $a_{i_j+1} \to \infty$, let $p_{i_j}/q_{i_j}$ be the corresponding convergents of $\alpha$, and set $t_j = \log q_{i_j}$.

From eq. (10), $T_j = g_{-t_j}T$ is defined by the lattice

$$\Lambda_{i_j} = \operatorname{span}\left\{\left(1, (-1)^{i_j} q_{i_j} \left\|q_{i_j}\alpha\right\|\right), \left(\frac{q_{i_j-1}}{q_{i_j}}, (-1)^{i_j-1} q_{i_j} \left\|q_{i_j-1}\alpha\right\|\right)\right\}.$$

Applying Lemma 4.3 along $\{t_j\}$ we obtain segments $I_j \subset T$ such that

$$M = T\#_{I_j}T, \quad \operatorname{hol}_T(I_j) = (x_j, y_j), \tag{28}$$

and

$$\frac{|x_j|}{q_{i_j}} + |y_j| q_{i_j} \to 0.$$

For each $j$, represent $M_j = g_{-t_j}M$ via the affine map associated with $g_{-t_j}$ as

$$M_j = T_j \#_{I'_j} T_j,$$

with

$$\operatorname{hol}_{T_j}(I'_j) = \left(\frac{x_j}{q_{i_j}},\, y_j q_{i_j}\right).$$

For $0 \le m \le \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor$, let $J_j^m \subset T_j$ satisfy

$$\operatorname{hol}_{T_j}(J_j^m) = \left(\frac{x_j}{q_{i_j}} + 2m,\, y_j q_{i_j}\right).$$

Then

$$\operatorname{Rel}_{2m} M_j = T_j \#_{J_j^m} T_j = T_j \#_{J'^m_j} T_j, \tag{29}$$

with

$$\operatorname{hol}_{T_j}(J'^m_j) = \left(\frac{x_j}{q_{i_j}},\, y_j q_{i_j} + (-1)^{i_j+1} 2m q_{i_j} \left\|q_{i_j}\alpha\right\|\right).$$

The first equality in eq. (29) follows by definition of the real Rel flow. The last equality follows from the first part of Proposition 6.6 and the fact that

$$\operatorname{hol}_{T_j}(J_j^m) - \operatorname{hol}_{T_j}(J'^m_j) \in 2\Lambda_{i_j}.$$

The segments $J_j^m$ and $J'^m_j$ share endpoints and intersect $2m-1$ times in their interiors at equally spaced points. By Proposition 6.6,

$$T_j \setminus \left(J_j^m \cup J'^m_j\right)$$

decomposes into $2m$ polygons, which can be two-colored so that adjacent polygons have different colors. Denote by $A_1$ and $A_2$ the unions of polygons of each color.

Among these, $2m-1$ are parallelograms with sides

$$\left(\frac{\frac{x_j}{q_{i_j}} + 2m}{2m}, \frac{y_j q_{i_j}}{2m}\right), \quad \left(\frac{\frac{x_j}{q_{i_j}}}{2m}, \frac{y_j q_{i_j} + (-1)^{i_j+1} 2m q_{i_j} \left\|q_{i_j}\alpha\right\|}{2m}\right).$$

Label them $p_1, \dots, p_{2m-1}$ starting from the marked point of $T_j$ in the direction of $J_j^m$.

The remaining polygon $p_{2m}$ has two vertices at the endpoints of $J_j^m$ (and $J'^m_j$) and shares edges with $p_1$ and $p_{2m-1}$.

Without loss of generality, assume that the parallelograms

$$p_1, p_3, \dots, p_{2m-1}$$

form $A_1$, and the remaining polygons

$$p_2, p_4, \dots, p_{2m}$$

form $A_2$.

For $k = 1, \dots, 2m-1$, we compute

$$\operatorname{area}(p_k) = \left| \det \begin{pmatrix} \frac{x_j}{2mq_{i_j}} + 1 & \frac{y_j q_{i_j}}{2m} \\ \frac{x_j}{2mq_{i_j}} & \frac{y_j q_{i_j}}{2m} + (-1)^{i_j+1} q_{i_j} \left\| q_{i_j} \alpha \right\| \end{pmatrix} \right|.$$

A direct computation yields

$$\operatorname{area}(p_k) = \left| \frac{y_j q_{i_j}}{2m} + (-1)^{i_j+1} \left( \frac{x_j}{2m} + q_{i_j} \right) \left\| q_{i_j} \alpha \right\| \right|, \quad k = 1, \dots, 2m-1.$$

The areas $A_1$ and $A_2$ depend on the integer $m$ and the subindex $i_j$. We avoid introducing this in the notation for convenience. Denote the area of the region $A_1$ by $b_j^m = \operatorname{area}(A_1)$, then

$$b_j^m = m \cdot \operatorname{area}(p_1) = \left| \frac{y_j q_{i_j} + (-1)^{i_j+1} x_j \left\| q_{i_j} \alpha \right\|}{2} + (-1)^{i_j+1} m q_{i_j} \left\| q_{i_j} \alpha \right\| \right|.$$

To estimate the values of $A_1$ for different $m$ and $j$, notice that

$$b_j^m = |\varepsilon_j + (-1)^{i_j+1} m \alpha_j|.$$

The value $\varepsilon_j \to 0$ as $j \to \infty$ independently of $m$,

$$\alpha_j = q_{i_j} \left\| q_{i_j} \alpha \right\| \in \left( \frac{1}{a_{i_j+1} + 1}, \frac{1}{a_{i_j+1}} \right)$$

is irrational and $m$ is a positive integer assuming the values

$$\left\{ 1, 2, \cdots, \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor \right\}.$$

**Claim.** *For every $b \in [0, 1/2]$ and every $\delta > 0$, for all sufficiently large $j$ there exists*

$$m \in \left\{ 1, \dots, \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor \right\}$$

*such that*

$$|b_j^m - b| < \delta.$$

We apply the claim it to finish the proof of Lemma 6.8 and finish with the proof of the claim at the end.

The claim implies that for all large $j$ the values of the area exchange

$$h_j^m = |\operatorname{area}(A_1) - \operatorname{area}(A_2)| = |1 - 2b_j^m|$$

become dense in $[0, 1]$ as $m$ varies. Therefore, given any fixed $h \in [0, 1]$, for all sufficiently large $j$ we can choose

$$m_j \in \left\{ 1, \dots, \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor \right\}$$

so that

$$h_j^{m_j} \to h \quad \text{as } j \to \infty.$$

Applying the affine map corresponding to $g_{t_j}$, we obtain two representations of $\operatorname{Rel}_{2m_j q_{i_j}} M$

$$T \#_{F_j^{m_j}} T = T \#_{F'^{m_j}_j} T,$$

where

$$\operatorname{hol}_T(F_j^{m_j}) = \left( x_j + (-1)^{i_j+1} 2 m_j q_{i_j},\, y_j \right) \tag{30}$$

and

$$\operatorname{hol}_T(F'^{m_j}_j) = \left( x_j,\, y_j + (-1)^{i_j+1} 2 m_j \left\| q_{i_j} \alpha \right\| \right). \tag{31}$$

The corresponding area exchange satisfies $h_j^{m_j} \to h$ as $j \to \infty$. We obtained the sequence $s_j = 2 m_j q_{i_j}$ such that

$$\operatorname{Rel}_{s_j} M \longrightarrow M \quad \text{as} \quad j \to \infty,$$

because the period coordinates of $M$ and $\operatorname{Rel}_{2m_j q_{i_j}} M$ inherited by $T$ are the same and we found representations as connected sums eqs. (28) and (31) with slits that have the same $x-$coordinate and differ on the $y-$coordinate by

$$0 < 2 m_j \left\| q_{i_j} \alpha \right\| \le a_{i_j+1} \left\| q_{i_j} \alpha \right\| < \frac{a_{i_j+1}}{q_{i_j+1}} \longrightarrow 0 \quad \text{as } j \to \infty.$$

*Proof of the claim.* Since

$$\alpha_j \in \left( \frac{1}{a_{i_j+1}+1}, \frac{1}{a_{i_j+1}} \right),$$

the values

$$\varepsilon_j + (-1)^{i_j+1} m\alpha_j, \qquad m = 1, \dots, \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor,$$

form an arithmetic progression with step size $\alpha_j \to 0$. Moreover,

$$\left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor \alpha_j \to \frac{1}{2},$$

and $\varepsilon_j \to 0$ as $j \to \infty$. Hence the set of values

$$\left\{ b_j^m : 1 \le m \le \left\lfloor \frac{a_{i_j+1}}{2} \right\rfloor \right\}$$

becomes dense in $[0, 1/2]$ as $j \to \infty$. △

□

We now deduce Proposition 6.4 from the previous result.

*Proof of Proposition 6.4.* Let $h \in [0,1]$ be fixed. Without loss of generality, assume that $T$ has area one and $M$ has area two.

By Lemma 6.8 and Theorem 4.5, there exists a sequence $s_j \to \infty$ such that $\mathrm{Rel}_{s_j} M \to M$. For each $j$, $M_j = \mathrm{Rel}_{s_j} M$ admits markings

$$T^1 \#_{F_j} T^2 = T^1 \#_{F_j'} T^2,$$

while $M$ admits markings

$$T^1 \#_{I_j} T^2,$$

with

$$\mathrm{hol}_{M_j}(F_j) = \mathrm{hol}_M(I_j) + (s_j, 0)$$

and

$$\left\| \mathrm{hol}_M(I_j) - \mathrm{hol}_{M_j}(F_j') \right\| \to 0. \tag{32}$$

Let $h_j$ denote the area exchange associated to $F_j$ and $F_j'$, with $h_j \to h$ as $j \to \infty$.

Let $\beta \in \mathcal{T}_M^{(0)}$ be fixed. By Lemma 4.2,

$$\beta = \frac{a}{2}(\beta^+ - \beta^-),$$

where $a > 0$ and $\beta^+, \beta^- \in C_M^{+,erg}$ are distinct with $\beta^+ = \iota(\beta^-)$ and $L_M(\beta^\pm) = 1$. Let $\mu^\pm$ be the corresponding ergodic invariant probability measures, with transverse measures $\nu^\pm$ and associated foliation cocycles $\beta^\pm$.

Let $\mu_j^+ = \mathrm{Leb}|_{A_j}$, where

$$A_j \subset T^1 \#_{I_j} T^2 \setminus (I_j \cup \iota(I_j))$$

is a connected component. For simplicity, assume $A_j = T^1 \setminus (I_j \cup \iota(I_j))$.

Then $\mu_j^+$ is invariant under the flow in the direction of $I_j$. Let $\nu_j^+$ be the associated transverse measure and $\beta_j^+ = \beta_{\nu_j^+}$ the corresponding cocycle. Define $\mu_j^+$, $\nu_j^-$ and $\beta_j^-$ by replacing $A_j$ with $\iota(A_j)$.

By Theorem 4.5, after passing to a subsequence we have

$$\beta_j^\pm \to \beta^\pm.$$

Let $B_j \subset T^1 \#_{F_j} T^2 \setminus (F_j \cup \iota(F_j))$ be a connected component, and set $\widehat{\mu}_j^+ = \mathrm{Leb}|_{B_j}$. For simplicity, assume that $B_j = T^1 \setminus (F_j \cup \iota(F_j))$.

For each $j$, the measure $\widehat{\mu}_j^+$ is invariant under the flow on $M_j$ in the direction of $F_j$. Define $\widehat{\mu}_j^- = \iota^*(\widehat{\mu}_j^+)$. Let $\widehat{\nu}_j^+$ and $\widehat{\nu}_j^-$ be the corresponding transverse measures.

Denote by $R_j\beta^+, R_j\beta^- \in C_{M_j}^{+,erg}$ the corresponding foliation cocycles on $M_j$ satisfying

$$\mathrm{trem}_{\frac{a}{2} R_j \beta^+} M_j = \mathrm{Rel}_{s_j} \mathrm{trem}_{\beta^+} M$$

and

$$\operatorname{trem}_{\frac{a}{2}R_j\beta^-} M_j = \operatorname{Rel}_{s_j} \operatorname{trem}_{\beta^-} M.$$

Let $C_j \subset T^1 \#_{F'_j} T^2 \setminus (F'_j \cup \iota(F'_j))$ be a connected component. After replacing $C_j$ by $\iota(C_j)$ if necessary, we may assume that

$$\widehat{\mu}_j^+(C_j) = \frac{1+h_j}{2} \quad \text{and} \quad \widehat{\mu}_j^+(\iota(C_j)) = \frac{1-h_j}{2}.$$

We represent the period coordinates of the marking of $M$ by

$$\{\xi_k^i : i,k \in \{1,2\}\} \cup \{I_j\},$$

where $\{\xi_k^i : k \in \{1,2\}\}$ is a basis of $H_1(T^i)$ for each $i \in \{1,2\}$ viewed as elements of relative homology $H_1(S,\Sigma)$. If $T$ is normalized as in Section 3.2, assume that

$$\operatorname{hol}_T(\xi_1^i) = (0,1) \quad \text{and} \quad \operatorname{hol}_T(\xi_2^i) = (1,\alpha)$$

for $i \in \{1,2\}$. To simplify notation, we write

$$\{\xi_k^i : i,k \in \{1,2\}\} \cup \{F'_j\}$$

for the period coordinates of the marking of $M_j$.

By definition of real Rel,

$$\operatorname{hol}_{M_j}(\xi_k^i) = \operatorname{hol}_M(\xi_k^i).$$

By the tremor comparison homeomorphism,

$$\operatorname{hol}_{\operatorname{trem}_{\frac{a}{2}(R_j\beta^+ - R_j\beta^-)} M_j}(\xi_k^i) = \operatorname{hol}_M(\xi_k^i) + \left(\frac{a}{2}(R_j\beta^+ - R_j\beta^-)(\xi_k^i), 0\right),$$

and

$$\operatorname{hol}_{\operatorname{trem}_{\frac{a}{2}(R_j\beta^+ - R_j\beta^-)} M_j}(F'_j) = \operatorname{hol}_{M_j}(F'_j) + \left(\frac{a}{2}(R_j\beta^+ - R_j\beta^-)(F'_j), 0\right).$$

By the bounds and limits

$$0 \le R_j\beta^\pm(F'_j) \le \operatorname{hol}_T^{(y)}(F'_j) \to 0, \qquad 0 \le \beta^\pm(I_j) \le \operatorname{hol}_T^{(y)}(I_j) \to 0,$$

together with eq. (32), it suffices to show that

$$R_j\beta^+ - R_j\beta^- \longrightarrow h(\beta^+ - \beta^-) \quad \text{as } j \to \infty,$$

which implies

$$\operatorname{Rel}_{s_j} \operatorname{trem}_\beta M \longrightarrow \operatorname{trem}_{h\beta} M \quad \text{as } j \to \infty.$$

The area exchange and the period coordinates of the marking $T^1 \#_{F'_j} T^2$ imply that

$$R_j\beta^\pm(\xi_k^i) - \beta_{\widehat{\nu}_j^\pm}(\xi_k^i) \longrightarrow 0 \quad \text{for all } i,k \in \{1,2\}.$$

The conclusion follows from the following claim.

**Claim.** *As $j \to \infty$,*

$$\widehat{\mu}_j^\pm|_{C_j} \xrightarrow{\text{weak}-*} \frac{1 \pm h}{2} \operatorname{Leb}_{T^1}, \quad \widehat{\mu}_j^\pm|_{\iota(C_j)} \xrightarrow{\text{weak}-*} \frac{1 \mp h}{2} \operatorname{Leb}_{T^2}.$$

We complete the proof and postpone the details of the claim to the end.

The claim implies that the foliation cocycles corresponding to the transverse measures $\widehat{\nu}_j^\pm$ satisfy

$$\beta_{\widehat{\nu}_j^+} \longrightarrow \frac{1+h}{2}\beta^+ + \frac{1-h}{2}\beta^-$$

and

$$\beta_{\widehat{\nu}_j^-} \longrightarrow \frac{1-h}{2}\beta^+ + \frac{1+h}{2}\beta^-.$$

Taking the difference, we obtain the desired limit

$$R_j\beta^+ - R_j\beta^- \longrightarrow h(\beta^+ - \beta^-) \quad \text{as } j \to \infty.$$

*Proof of the claim.* The argument follows [CSW25, Lemma 10.7]. It suffices to show that

$$\frac{2}{1+h_j}\,\widehat{\mu}_j^+|_{C_j} \xrightarrow{\text{weak}-*} \mathrm{Leb}_{T^1},$$

since $h_j \to h$ implies

$$\widehat{\mu}_j^+|_{C_j} \xrightarrow{\text{weak}-*} \frac{1+h}{2}\,\mathrm{Leb}_{T^1}.$$

The remaining statements follow analogously.

The horizontal flow on $T^1$ is uniquely ergodic; in particular, $\mathrm{Leb}_{T^1}$ is the unique invariant probability measure on $T^1$. Denote by $\Upsilon_j(x,t)$ the image of $x \in T^1$ under the straight-line flow in the direction of $F_j$ at time $t$. Similarly, denote by $\Upsilon(x,t)$ the image of $x$ under the horizontal flow at time $t$. The measure

$$\frac{2}{1+h_j}\,\widehat{\mu}_j^+|_{C_j}$$

is a convex combination of length measures supported on segments of the form

$$\{\Upsilon_j(x,t) : t \in [0, S(j)]\}, \quad x \in F_j',$$

where $S(j)$ is the first return time to $F_j'$, independent of $x$. As $F_j$ and $F_j'$ become longer and increasingly horizontal, we have $S(j) \to \infty$.

By unique ergodicity, for every continuous function $f$ on $T^1$ and every $\varepsilon > 0$, there exists $S_0$ such that for all $S \geq S_0$,

$$\left| \frac{1}{S}\int_0^S f(\Upsilon(x,t))\ \mathrm{d}t - \int f\ \mathrm{d}\,\mathrm{Leb}_{T^1} \right| < \frac{\varepsilon}{2},$$

uniformly in $x \in T^1$.

By uniform continuity of $f$, for every fixed $S$ and all $j$ sufficiently large,

$$\left| \frac{1}{S}\int_0^S f(\Upsilon(x,t))\ \mathrm{d}t - \frac{1}{S}\int_0^S f(\Upsilon_j(x,t))\ \mathrm{d}t \right| < \frac{\varepsilon}{2}.$$

By the triangle inequality, we obtain the desired convergence. △

□

## References


[AM24] Jayadev S. Athreya and Howard Masur, *Translation surfaces*, Graduate Studies in Mathematics, vol. 242, American Mathematical Society, Providence, RI, [2024] ©2024. MR 4783430

[BSW22] Matt Bainbridge, John Smillie, and Barak Weiss, *Horocycle dynamics: new invariants and eigenform loci in the stratum* $\mathcal{H}(1,1)$, Mem. Amer. Math. Soc. **280** (2022), no. 1384, v+100. MR 4493581

[CDF23] Gabriel Calsamiglia, Bertrand Deroin, and Stefano Francaviglia, *A transfer principle: from periods to isoperiodic foliations*, Geom. Funct. Anal. **33** (2023), no. 1, 57–169. MR 4561148

[CE07] Yitwah Cheung and Alex Eskin, *Slow divergence and unique ergodicity*, 2007.

[CSW25] Jon Chaika, John Smillie, and Barak Weiss, *Tremors and horocycle dynamics on the moduli space of translation surfaces*, Annals of Mathematics (2025), forthcoming.

[CW10] Kariane Calta and Kevin Wortman, *On unipotent flows in* $\mathcal{H}(1,1)$, Ergodic Theory Dynam. Systems **30** (2010), no. 2, 379–398. MR 2599885

[CW24] Jon Chaika and Barak Weiss, *On the ergodic theory of the real rel foliation*, Forum of Mathematics, Pi **12** (2024), e7.

[EMS03] Alex Eskin, Howard Masur, and Martin Schmoll, *Billiards in rectangles with barriers*, Duke Math. J. **118** (2003), no. 3, 427–463. MR 1983037

[FM12] Benson Farb and Dan Margalit, *A primer on mapping class groups (pms-49)*, Princeton University Press, 2012.

[Ham18] Ursula Hamenstädt, *Ergodicity of the absolute period foliation*, Israel J. Math. **225** (2018), no. 2, 661–680. MR 3805661

[Hil61] Furstenberg Hillel, *Strict ergodicity and transformation of the torus*, American Journal of Mathematics **83** (1961), no. 4, 573–601. MR 133429

[HW18] W. Patrick Hooper and Barak Weiss, *Rel leaves of the Arnoux-Yoccoz surfaces*, Selecta Math. (N.S.) **24** (2018), no. 2, 875–934, With an appendix by Lior Bary-Soroker, Mark Shusterman, and Umberto Zannier. MR 3782414

[KMS86] Steven P. Kerckhoff, Howard Masur, and John Smillie, *Ergodicity of billiard flows and quadratic differentials*, Annals of Mathematics **124** (1986), no. 2, 293–311, (2).

[McM07a] Curtis T. McMullen, *Dynamics of* $\mathrm{SL}_2(\mathbb{R})$ *over moduli space in genus two*, Ann. of Math. (2) **165** (2007), no. 2, 397–456. MR 2299738

[McM07b] ———, *Foliations of Hilbert modular surfaces*, Amer. J. Math. **129** (2007), no. 1, 183–215. MR 2288740
[McM13a] ———, *Diophantine and ergodic foliations on surfaces*, Journal of Topology **6** (2013), no. 2, 349–360. MR 3065179
[McM13b] ———, *Navigating moduli space with complex twists*, J. Eur. Math. Soc. (JEMS) **15** (2013), no. 4, 1223–1243. MR 3055760
[McM14] ———, *Moduli spaces of isoperiodic forms on Riemann surfaces*, Duke Math. J. **163** (2014), no. 12, 2271–2323. MR 3263035
[MS91] Howard Masur and John Smillie, *Hausdorff dimension of sets of nonergodic measured foliations*, Annals of Mathematics **134** (1991), no. 3, 455–543. MR 1135877
[MT02] Howard Masur and Serge Tabachnikov, *Rational billiards and flat structures*, Handbook of dynamical systems, Vol. 1A, North-Holland, Amsterdam, 2002, pp. 1015–1089. MR 1928530
[MW14] Yair Minsky and Barak Weiss, *Cohomology classes represented by measured foliations, and Mahler's question for interval exchanges*, Ann. Sci. Éc. Norm. Supér. (4) **47** (2014), no. 2, 245–284. MR 3215923
[Osp24] Carlos Ospina, *Some real rel trajectories in $\mathcal{H}(1,1)$ that are not recurrent*, 2024.
[Sch05] Martin Schmoll, *Spaces of elliptic differentials*, Algebraic and topological dynamics, Contemp. Math., vol. 385, Amer. Math. Soc., Providence, RI, 2005, pp. 303–320. MR 2180242
[Vee69] William A. Veech, *Strict ergodicity in zero dimensional dynamical systems and the kronecker-weyl theorem mod 2*, Transactions of the American Mathematical Society **140** (1969), 1–33. MR 240056
[Win22] Karl Winsor, *Dense real rel flow orbits and absolute period leaves*, 2022.
[Wri15] Alex Wright, *Cylinder deformations in orbit closures of translation surfaces*, Geom. Topol. **19** (2015), no. 1, 413–438. MR 3318755
[Ygo22] Florent Ygouf, *Isoperiodic dynamics in rank 1 affine invariant orbifolds*, International Mathematics Research Notices **2022** (2022), no. 24, 19854–19861.

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